



# UNIVERSITÀ DI PISA

Department of Computer Science  
Bachelor's Degree in Computer Science

BACHELOR'S THESIS

**Can we recover differential transcript usage  
(DTU) from scRNA-seq?**

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# Abstract

Single-cell RNA sequencing has transformed the study of cellular heterogeneity, yet widely adopted droplet-based protocols capture only the 3' end of each transcript [2]. This constraint makes Differential Transcript Usage (DTU) difficult to detect, as short terminal reads lack the internal splice junctions required to distinguish isoforms. Furthermore, proportion-based DTU methods—including those extended to single-cell data, which rely on pseudo-bulk aggregation—sacrifice cell-level resolution to recover the read depth they require [3, 4].

This thesis investigates whether COTAN—a statistical framework that models zero counts directly through presence/absence contingency tables [1]—can be extended from gene-level co-expression to transcript-level DTU detection. By bypassing both artificial imputation and pseudo-bulk aggregation, the adapted method identifies mutually exclusive isoform pairs via independence testing and bidirectional contrast thresholds across cell clusters.

A reproducible Nextflow pipeline was built to construct transcript-level matrices from raw 3'-biased data using alevin-fry with identity transcript-to-gene mapping [5]. This pipeline was applied to a human cell line mixture (29,000 cells) and mouse brain tissue (4,000 cells). Transcript-level clustering successfully recovered fine-grained substructures consistent with the Global Differentiation Index, and top DTU candidates—including *Meg3*, *Malat1*, and *TPM1*—were consistent with documented isoform biology [6].

These results demonstrate that sparsity-robust, contingency-table statistics can detect isoform mutual exclusivity in droplet-based data without the prohibitive cost of full-length sequencing. While sensitivity remains bounded by positional bias—rendering mid-transcript splicing events invisible—and orthogonal validation is still required, this work establishes a principled path for isoform-aware analysis on large-scale atlas datasets.



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# Glossary

This glossary defines the core concepts whose discussion spans several chapters. Terms needed only once are defined inline as footnotes where they first appear; the entries below are those the reader may want to revisit. Each term is set in italics in the text, so a reader can recognise it as an entry defined here.

**3' end** The terminal end of an RNA molecule at which transcription finishes; in mRNA, it is where the poly-A tail is added. In 3'-biased protocols, the capture chemistry is anchored at this end, so reads cover only the terminal fragment of each molecule rather than the whole transcript. 3, 34

**3'-biased sequencing** A property of some scRNA-seq protocols (such as 10x Genomics Chromium) in which sequencing reads are derived only from the 3' terminus of each transcript. Because reading quality decreases by distance and because capture is anchored at the transcript's 3' end, coverage does not extend into the transcript body, so internal splice junctions are never observed [7]. 3

**alternative splicing** The process by which a gene's long primary RNA copy is shortened and rearranged during maturation. The copy contains coding blocks (exons) and non-coding blocks (introns); splicing removes the introns and joins the flanking exons at the resulting splice junctions. When exon selection varies between transcripts of the same gene, the result is multiple distinct isoforms. 1

**bulk RNA-seq** Sequencing of RNA pooled from many cells at once, which averages away cell-to-cell differences. 2, 3, 12

**cell barcode** A short DNA tag carried by every read from a given cell, identifying its cell of origin. It allows the reads of one pooled sequencing run to be regrouped into per-cell expression profiles. 2, 7, 11

**co-expression** The tendency of two features (genes or transcripts) to be detected together across cells. 22, 24

**contingency table** A  $2 \times 2$  table counting, across all cells, the four presence/absence combinations of two features; it tests whether they co-occur more or less than chance. 12, 24

**count matrix** A table whose rows index features (genes or transcripts) and whose columns index cells, with each entry recording how many molecules of that feature were detected in that cell. 3, 4, 6

**differential gene expression (DGE)** The analysis of differences in a gene's total expression—the summed expression of all its isoforms—between groups of samples. It measures overall gene output and is blind to changes in the relative usage of individual isoforms, which are instead captured by differential transcript usage. 3, 14

**Differential Transcript Usage (DTU)** Differential transcript usage occurs when the relative expression of multiple transcripts arising from the same gene changes between different conditions [8]. It is independent of changes in the gene’s total expression, so it is not captured by gene-level differential expression analysis and requires transcript-level resolution to detect. 1, 3, 4, 14, 17, 20, 25, 35

**droplet-based protocols (droplet data)** High-throughput single-cell protocols that encapsulate each cell together with a barcoded bead inside a picolitre droplet, so that tens of thousands of cells are processed in one run. They are the dominant scRNA-seq format, and their capture chemistry reads only the tail of each transcript. 2

**FASTQ** The plain-text de facto standard output of Illumina sequencers, named by analogy to FASTA with a Q for quality. Files are almost always gzip-compressed, taking the `.fastq.gz` extension. 7, 17, 18

**full-length sequencing** A sequencing approach that reads an RNA molecule from one end to the other, rather than only the fragment nearest one of its ends. 2

**gene** A functional unit of the genome that specifies one or more RNA molecules. 1, 18

**genome** The complete set of genetic material of an organism, written in DNA and inherited from one generation to the next. It is organised into genes, the functional units whose expression is measured. 1

**Global Differentiation Index (GDI)** A cluster-specificity measure used to identify marker genes and validate clustering quality. 13, 23, 25, 26, 35, 36

**isoform** A distinct mature transcript variant produced from the same gene by alternative splicing. 1, 3, 4, 7

**isoform switching** A change between conditions in the relative usage of a gene’s isoforms, such that the fraction of the gene’s expression contributed by one isoform increases while that of another decreases. It reflects altered alternative splicing or transcription and is quantified by the change in each isoform’s fraction of total gene expression [9]. 3, 15

**messenger RNA (mRNA)** The protein-coding subtype of RNA that carries a gene’s sequence to the ribosome. 1

**mutual exclusivity** Two features that rarely appear together in the same cell (a negative association), as opposed to co-expression. 15, 24

**PCR** Polymerase chain reaction: the amplification step that copies each captured cDNA fragment many times over, so that the few molecules recovered from a cell are present in enough quantity for the sequencer to read [10]. Because copies outnumber original molecules, the UMI tag is what allows them to be collapsed back to a single count. 2, 6, 7, 20

**protein** A molecule built from a chain of amino acids that performs the work of the cell, including catalysing reactions, transmitting signals, and forming structural components. 1

**pseudo-bulk aggregation** The practice of summing read counts across groups of single cells to simulate bulk RNA-seq depth, which recovers statistical power at the expense of single-cell heterogeneity. 3

**read** A short nucleotide sequence produced by the sequencing machine, representing the raw data of RNA-seq. 2

**RNA (ribonucleic acid)** The broad class of nucleic acid molecules transcribed from DNA. It comprises messenger RNA (mRNA), which carries protein-coding instructions, and non-coding RNAs that perform structural or regulatory roles. 1

**RNA sequencing (RNA-seq)** A technique that determines the sequence and abundance of RNA molecules in a sample; it is the basis of the single-cell protocols discussed in this thesis. 2

**single-cell RNA sequencing (scRNA-seq)** RNA sequencing performed on thousands of individual cells separately, exposing per-cell expression. 2, 6, 18

**sparsity** The very high proportion of zero counts in single-cell RNA-seq data. 6

**splice junction** The boundary created where two coding blocks (exons) are joined after the intervening non-coding block (intron) has been removed. Reads spanning a splice junction identify which blocks were joined together. 1

**tag bias** A bias in which each molecule is read only from one of its ends, so that coverage concentrates near that end while the interior of the transcript is never observed. It follows from a capture chemistry anchored at a single end, as in the tail-primed droplet protocols. 2, 3, 4, 12

**technical dropout** A zero count produced by a technical limitation of the assay—failed capture, reverse transcription, or sequencing—rather than by the true absence of the transcript. It is also described as a non-biological zero. 11, 14

**transcript** An RNA molecule copied from a gene. 1, 4, 6

**Unique Molecular Identifier (UMI)** A short random sequence attached to each captured RNA molecule before PCR amplification, so that amplified copies of one molecule are recognised and counted only once. 2, 7, 11, 19, 20

**unspliced/spliced transcript** The first RNA copy of a gene, still containing its non-coding blocks (introns); it is also called the primary transcript or pre-mRNA. Its counterpart is the spliced transcript, the same molecule once those blocks have been removed, which is the mature form translated into protein. 1



# Chapter 1

## Introduction

### Biological Context for Computer Scientists

*Proteins* are the basic building blocks of a cell and its DNA *genome* encodes the information needed to build them. Every cell of a multicellular organism carries essentially the same DNA, yet cells develop into distinct types. The difference cannot come from the DNA itself. Between the reading of the genome and the building of proteins, therefore, lies one or more layers of regulation that determine which proteins a cell produces, and those proteins in turn give the cell its characteristics. Studying these layers is central to explaining how cells function and what shapes their behaviour.

Gene expression proceeds in three stages. A *gene* is first copied into a long molecule of RNA (*ribonucleic acid*), called *unspliced transcript*. The second stage, *alternative splicing*, removes the non-coding blocks and joins the coding ones at the resulting *splice junctions*, yielding a shorter spliced *transcript*. The final, spliced version is *messenger RNA (mRNA)*, which the ribosome translates<sup>1</sup> into protein in the third and final phase.

The shortening process is not a one-to-one mapping. To expand the decoding capacity of a gene, cells can produce several different spliced versions of the same *unspliced transcript*; these versions are called *isoforms*. Because different isoforms can translate into proteins with entirely different, or even conflicting, roles, a cell can fundamentally alter its behaviour simply by switching which variant it expresses—the MET $\Delta$ <sub>14</sub> isoform, for instance, acts as a switch that drives invasive growth in cancer [11]. The core goal of *Differential Transcript Usage (DTU)* analysis is therefore to measure changes in the relative expression proportions of specific transcript isoforms from the same gene across different biological conditions, developmental stages, or cell types, in order to expose regulatory events that gene-level measurement renders invisible.

<sup>1</sup> **Translation:** The process by which the ribosome reads an mRNA molecule and assembles the corresponding chain of amino acids into a protein.

## A Brief History of Single-Cell Sequencing

For most of its history, *RNA sequencing (RNA-seq)* was performed on bulk samples. RNA is extracted from a population of cells, pooled into a single reaction, and sequenced as one library<sup>2</sup>, so the resulting profile describes the sample as a whole rather than any cell within it. Sequencing of this kind, termed *bulk RNA-seq*, is statistically comfortable: pooling thousands to millions of cells yields deep coverage and stable abundance estimates for every gene. The averaging, however, is also its principal limitation. A gene's measured abundance is the mean over all cells present, so cellular heterogeneity<sup>3</sup> disappears before the data ever reach an analyst. A rare cell type constituting two percent of a tissue contributes two percent of that tissue's *reads* for each of its markers<sup>4</sup> and is easily buried beneath the dominant population; two subpopulations that shift in opposite directions—one gaining a transcript while the other loses it—cancel out and appear unchanged.

*Single-cell RNA sequencing (scRNA-seq)* replaced that aggregate view with per-cell profiling, running the same measurement independently on thousands of cells. The strategy rests on compartmentalisation. Each cell is physically isolated, its polyadenylated RNA captured and reverse-transcribed separately, and every resulting molecule marked with two labels. A *cell barcode* identifies the cell of origin, so reads can be regrouped into per-cell profiles after sequencing; a *Unique Molecular Identifier (UMI)* tags the individual molecule, so *PCR (polymerase chain reaction)* duplicates are collapsed rather than miscounted as expression. A single pooled sequencing run is thereby converted into thousands of independent expression measurements, each far shallower than a bulk sample.

The first *scRNA-seq*, produced from 2009 onwards, came from plate-based protocols, which consume one cell per reaction [12]. The chemistry of that era anchored its reads at one end of the molecule, and base accuracy declines along a read as it lengthens.<sup>5</sup> Reliable information therefore concentrated near that end, a limitation that deeper sequencing could not remedy—data of this kind exhibit *tag bias* [13].

In 2012, the invention of *full-length sequencing* removed that bias [14, 15]. A whole molecule could now be copied and read from end to end, preserving the internal structure that distinguishes one transcript variant from another. While this was a substantial gain in coverage, the technology was costly and the plate format kept a study at hundreds to a few thousand cells.

A new protocol, droplet microfluidics, suddenly revolutionised the scale in 2015 [16]: each droplet encapsulates one cell together with a bead<sup>6</sup> carrying barcoded primers<sup>7</sup>, so tens of thousands of cells are profiled in one run at a small fraction of the cost per cell. The gain came at a price. Droplet capture is anchored at the transcript tail; this forces the use of the old chemistry, and the tag bias returned.

*Droplet protocols* dominate single-cell work today, since they profile far more cells per experiment at a far lower cost per cell than plate-based chemistry allows [2]. What tag bias removes is therefore the central limitation of the format rather than an incidental one, and recovering it is an active line

<sup>2</sup> **Sequencing library:** The set of DNA molecules loaded into the sequencer for one run. Here it holds the sample's RNA, copied into DNA and tagged so that the instrument can read each fragment.

<sup>3</sup> **Cellular heterogeneity:** The coexistence of distinct cell types and states within a single organism.

<sup>4</sup> **Marker genes:** A gene or transcript whose expression specifically identifies a cell type, state, or condition.

<sup>5</sup> **Base:** One of the four nucleotides that form a DNA or RNA strand: adenine (A), cytosine (C), guanine (G), and thymine (T). A read is a sequence of consecutive bases, and the confidence with which each base is identified declines along a read as it lengthens.

<sup>6</sup> **Bead:** A small gel sphere loaded with the barcoded primers. One bead is co-encapsulated with each cell, supplying every primer its droplet needs.

<sup>7</sup> **Barcoded primers:** Primers that carry a short identifying sequence, so that every molecule captured in a given droplet can be traced back to its cell after sequencing.

of work. Long-read platforms read whole transcripts at single-cell resolution, and short-read datasets are mined for whatever isoform signal their terminal fragments still carry [6].

## The Computational Challenge

Turning droplet data into transcript-level results means working around two limitations that the sequencing chemistry itself imposes, one on the quantity of data and one on its content. The first is sparsity. *Count matrices* from droplet experiments are mostly zeros, with zero entries routinely forming 70% to 90% of all values. Some of those zeros are biological, recording a transcript that the cell never produced, while others are technical, recording a molecule that was present but never captured, and both look identical in the matrix.

The second limitation is positional. Only the terminal 3' end of each molecule is read, so where several *isoforms* share that end they look identical in the data and their proportions cannot be measured directly. Deeper sequencing reduces the first limitation and cannot touch the second: extra reads fill in more of the matrix, but no quantity of terminal fragments recovers the interior of a transcript. A method that works on droplet data must therefore extract signal from a lightly observed count matrix while accepting that most of each molecule was never seen.

### A Note on What Remains Recoverable

Gene-level information survives the *tag bias* constraint. A gene registers when a read is confidently attributed to it, and in most cases a read that cannot be traced to a single transcript still maps to isoforms of the same gene. A gene matrix can therefore be built from the same reads, and it remains informative for *differential gene expression (DGE)* analysis. A consequential property is that this matrix is denser than the transcript matrix.

## Current Approaches and Limitations

Transcript-level regulation has not gone unstudied; the question is whether existing tools survive the sparsity and positional bias that droplet data impose. Three families of methods address it, and each stops short of detecting *DTU* across a full droplet dataset.

**Bulk algorithms.** Established *bulk RNA-seq* tools such as rMATS [17], SUPPA2 [18], DRIMSeq [19] and DEXSeq [20] model continuous isoform proportions from deep, junction-spanning coverage. Applied to sparse droplet matrices, those assumptions break down and the resulting estimates are unstable.

**Aggregation to pseudo-bulk.** One route around the coverage problem is *pseudo-bulk aggregation*, which sums the cells of a cluster or condition into a single profile to recover the depth that bulk tools expect [3, 21]. The cost is the cell-to-cell variation the assay was built to measure: a switch confined to a small subpopulation is diluted by the cells around it.

**Targeted 3'-biased sequencing tools.** Sierra [4] analyses droplet data directly, but only for a narrow class of events at transcript termini. It offers no genome-wide framework for *multi-isoform switching* from raw zero-heavy counts.

What is missing is a framework that works on sparse droplet matrices at single-cell resolution, without collapsing cells together and without full-length reads.

## Thesis Objective

The question this thesis pursues originates in COTAN (Co-expression Tables Analysis) [1], a framework for gene-level *count matrices* that focuses its statistical model on the zeros of the matrix rather than on its positive counts. COTAN asks whether two genes are detected together less often than chance would predict, using presence/absence contingency tables, so no artificial imputation<sup>8</sup> or variance-stabilising transformation<sup>9</sup> is needed, and it performs well on the sparse matrices typical of single-cell data. The present thesis asks whether the same statistics hold when applied one level lower, to *transcript* matrices whose counts are sparser still: whether they retain enough *isoform* information to detect *DTU* without fixing the *tag bias* problem. The claim is deliberately modest — isoforms that share their terminal region remain indistinguishable — and the aim is to map where sparsity-robust statistics recover differential transcript usage anyway.

The work proceeded from raw reads to a reported result in four stages, each producing a concrete artefact: a count-matrix pipeline, an analysis library, a reproducible driver layer, and a set of reported results.

**From reads to transcript matrices.** A Nextflow pipeline was built that turns raw droplet reads into a filtered transcript-level count matrix—the artefact the whole study rests on, since DTU analysis requires transcript resolution that gene-level matrices do not carry.

**An analysis library.** The analysis was consolidated into a single R package, *cotanisoform*, which extends COTAN’s gene-level framework to transcript resolution. It keeps COTAN’s statistics—co-expression, p-values, GDI, differential expression—but adds the steps a transcript-level analysis needs. It is where COTAN’s gene-framed statistics become a transcript-level detector.

**The analysis drivers.** A configuration-driven layer reproduces a full per-dataset analysis from a single declarative file, joining the pipeline and the library so the whole study runs end to end reproducibly.

**From candidates to reported results.** The workflow was applied to two datasets—a homogeneous human cell-line mixture and complex mouse cortical tissue—and the candidates are cross-referenced against published isoform biology and gene-level clustering baselines. The analysis was carried out first-hand and not reviewed by a domain expert; the findings are a reproducible first report rather than established biology, and independent verification is expected before they are built upon.

<sup>8</sup> **Imputation:** Replacing zero counts with values predicted from neighbouring cells, at the risk of inventing spurious correlations.

<sup>9</sup> **Variance-stabilising transformation:** A transformation applied to counts, such as a logarithm, so that their variance no longer depends on their mean. It alters the scale of the data, turning an undetected molecule into a small positive value rather than a zero.

## Chapter 2

# Theory and Frameworks

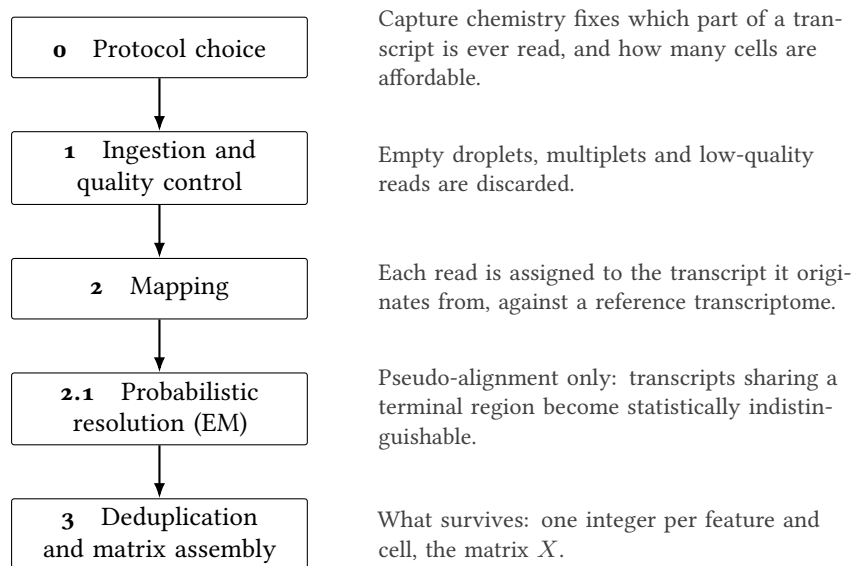
### Notation

| Symbol                       | Meaning                                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X$                          | Expression count matrix produced by the preprocessing pipeline, with features (transcripts or genes, depending on the pipeline output level) in rows and cells in columns |
| $x_{fc}$                     | Observed integer count of feature $f$ in cell $c$                                                                                                                         |
| $f, c$                       | Feature and cell indices                                                                                                                                                  |
| $N$                          | Number of cells                                                                                                                                                           |
| $R$                          | Total number of sequencing reads                                                                                                                                          |
| $L$                          | Sequenced read length                                                                                                                                                     |
| $M$                          | Mean number of reads per cell                                                                                                                                             |
| $k$                          | Length of the $k$ -mers used to index the reference                                                                                                                       |
| $ \Sigma $                   | Alphabet size (4 for DNA)                                                                                                                                                 |
| $T$                          | Number of annotated transcripts in the reference                                                                                                                          |
| $T_{obs}$                    | Distinct transcripts detected in a single cell                                                                                                                            |
| $L_{avg}$                    | Mean transcript length                                                                                                                                                    |
| $\theta_t$                   | Relative abundance of transcript $t$                                                                                                                                      |
| $\ell_t$                     | Effective length of transcript $t$                                                                                                                                        |
| $C_r$                        | Equivalence class of transcripts compatible with read $r$                                                                                                                 |
| $\hat{z}_{rt}$               | Fractional assignment of read $r$ to transcript $t$                                                                                                                       |
| <b>Statistics</b>            |                                                                                                                                                                           |
| $a_c, b_c$                   | Binary presence indicators for a pair of features in cell $c$                                                                                                             |
| $O_{ij}, E_{ij}$             | Observed and expected counts in a $2 \times 2$ contingency table                                                                                                          |
| $\text{Row}_i, \text{Col}_j$ | Marginal totals of that table                                                                                                                                             |
| $\chi^2$                     | Pearson chi-squared statistic                                                                                                                                             |
| $C_{f,f'}$                   | Co-expression coefficient between features $f$ and $f'$                                                                                                                   |
| $p_{f,f'}$                   | Independence $p$ -value for that pair                                                                                                                                     |
| <b>Detection</b>             |                                                                                                                                                                           |
| $D_{f,S}$                    | Differential expression (DEA) coefficient of feature $f$ in cluster $S$                                                                                                   |
| $S$                          | Cell cluster                                                                                                                                                              |
| $\tau$                       | Contrast threshold on the DEA difference within a gene                                                                                                                    |
| $\text{gdi}_i$               | Cluster-specificity index of feature $i$                                                                                                                                  |

## Preprocessing Pipeline

A preprocessing pipeline carries a single *scRNA-seq* experiment from its data-generating mechanism to the *count matrix* consumed by every later analysis. That transformation is the bridge between raw sequencing output and statistical input, and it is organised as three successive steps: ingestion and quality control, mapping, and deduplication into a matrix. Ingestion and quality control turn raw reads into clean units, discarding the failures of the chemistry—empty droplets, multiplets, and low-quality molecules—so that the cells entering the analysis are genuine and distinct. Mapping assigns each read to the *transcript* it originates from by comparing it against a reference transcriptome and reporting the set of transcripts with which it is compatible. Deduplication collapses the *PCR* copies that descend from one captured molecule, grouping reads by cell barcode, UMI, and assigned transcript, so the count matrix records the number of distinct molecules per cell and feature rather than the number of reads.

Furthermore, the protocol chosen at the outset is not an incidental detail: it sets the depth, resolution, and sparsity of the whole experiment, and a different chemistry would reshape every phase and the matrix it ultimately produces. The subsections that follow take up each step in order, following a single experiment as it moves from raw sequencing output to the count matrix that every later analysis consumes. Figure 2.1 summarises the four phases and records what each one costs.



**Figure 2.1:** The four phases of the preprocessing pipeline and the information each one removes. Phase 2.1 exists only on the pseudo-alignment route, where ambiguous reads must be distributed probabilistically; direct alignment resolves every read by its coordinates and skips it. The chemistry chosen in Phase 0 bounds every later stage, so the losses accumulate rather than cancel.

### Phase 0: Choosing the Protocol

The design of any *scRNA-seq* analysis pipeline is inherently constrained by the physical protocol used for data generation. This experimental choice dictates the scale, resolution, and *sparsity* of the resulting *count matrix*. Table 2.2 compares the droplet protocol against plate-based methods.

The widespread adoption of droplet protocols is ultimately driven by economics. While a plate-based experiment costs one to five dollars per cell, droplet encapsulation profiles cells for a few cents, allowing researchers to assay two to three orders of magnitude more cells on a fixed budget [2].

**Table 2.2:** The two dominant single-cell library chemistries. Per-cell depth and cost trade off against throughput; total reads per day are set by the sequencer, not the chemistry. All figures describe a single sequencing run. Values are order-of-magnitude, after Ding et al. [22].

|                               | Plate (SMART-seq2)    | Droplet (10x)                     |
|-------------------------------|-----------------------|-----------------------------------|
| Library preparation           | ~ 2 days for 96 cells | ~ 1 day for $10^4$ – $10^5$ cells |
| Cells per experiment          | $10^2$ – $10^3$       | $10^4$ – $10^5$                   |
| Reads per cell                | ~ $10^6$              | ~ $10^4$                          |
| Detected transcripts per cell | ~ $10^4$ – $10^5$     | ~ $10^3$                          |
| Reads per day                 | ~ $10^8$ – $10^9$     | ~ $10^8$ – $10^9$                 |
| Coverage                      | full transcript       | 3' terminus only                  |
| Reagent cost per cell         | ~ 10–100× higher      | baseline                          |

This trade-off between per-cell depth and cell quantity is highly advantageous for standard tasks like cell-type clustering, marker discovery, and differential gene expression. For these gene-level questions, the statistical power gained by averaging over thousands of cells overcomes the severe sparsity within individual cells. However, the loss of transcript coverage remains a fundamental limitation for *isoform*-level analyses.

## Phase 1: Data Ingestion and Quality Control

The first computational challenge in a single-cell pipeline is purely one of scale and structure. Sequencers output raw data in *FASTQ* format, representing each read across four distinct lines: a coordinate identifier<sup>1</sup>, the nucleotide sequence<sup>2</sup>, a separator<sup>3</sup>, and the corresponding Phred-scale quality scores<sup>4</sup>. For droplet-based chemistries such as the 10x Genomics platform, this raw sequence carries critical multiplexing metadata that must be carefully isolated before any biological analysis begins. Specifically, the sequencer must extract a *cell barcode* that maps the read back to its discrete cell of origin, and a *UMI*—a random tag attached prior to *PCR* amplification used to track individual captured mRNA molecules.

Before the reads can be quantified, they must be cleaned. Tools such as *cutadapt*<sup>5</sup> or *fastp*<sup>6</sup> trim synthetic adapters, discard low-quality bases, and extract the barcode and UMI payloads, while heuristic filtering separates valid cell barcodes from empty droplets and multi-cell collisions (multiplets). The step is optional, and is skipped under two conditions. When a vendor pipeline such as Cell Ranger has already trimmed adapters and corrected barcodes and UMIs, repeating the pass costs a full read of the *FASTQ* file without altering a base. When the mapper is read-structure-aware and UMI-aware, as *alevin-fry* and *Salmon* are, residual adapters fall outside the indexed *k*-mers and are never matched, so correction is left to the quantifier. Skipping is safe only under those guarantees: on untrimmed reads handled by a mapper that does not correct UMIs, adapters mis-assign reads and sequencing errors inflate the number of distinct UMIs. For plate-based chemistry no barcode or UMI exists to extract and adapter content is low, so trimming reduces to a quality filter.

<sup>1</sup> **Read identifier:** A line beginning with @ that names the read and records the instrument, flow cell and coordinates it came from.

<sup>2</sup> **Nucleotide sequence:** The read itself, a string of A, C, G and T bases in the order the sequencer read them.

<sup>3</sup> **Separator:** A line beginning with + that marks the end of the sequence and may repeat the identifier.

<sup>4</sup> **Per-base quality score:** One character per sequenced base, encoding the probability that the base was called incorrectly on the Phred scale.

<sup>5</sup> **cutadapt:** A read-trimming tool that removes adapter sequences and low-quality bases.

<sup>6</sup> **fastp:** A fast all-in-one read preprocessor that performs adapter trimming, quality filtering and reporting in a single pass.

Because quality control touches every single nucleotide exactly once, it acts as a highly parallelisable streaming operation strictly bound by the total read count,  $R$ , and the read length,  $L$ . The computational time complexity is entirely linear. The physical footprint, by contrast, is large. One uncompressed FASTQ record—a 100 bp read together with its identifier and quality line—occupies roughly 250 bytes, so a droplet experiment of  $R \sim 10^9$  reads, the order of magnitude considered here, holds  $\sim 2.5 \times 10^{11}$  bytes, about 250 GB before compression. Modern multi-core trimmers sustain throughputs on the order of  $10^8$  reads per minute, so this I/O-bound filtering resolves in a few hours on a single workstation.

## Phase 2: Mapping Algorithms (Alignment vs. Pseudo-Alignment)

Quantifying transcript expression requires mapping short sequencing reads back to their reference transcripts of origin. The computational workload is primarily driven by the total read count,  $R = N \cdot M$ . For the two datasets analysed in this thesis,  $R$  is approximately  $10^8$  to  $10^9$ , establishing  $10^9$  reads as a baseline scale (though pooled multi-sample atlases can easily reach  $10^{11}$  reads).

The mapping process is further defined by three variables: the number of annotated isoforms ( $T \sim 10^5$ ), the average transcript length ( $L_{avg} \approx 1500$  bp), and the sequenced read length ( $L \approx 90\text{--}150$  bp). In a 3' droplet assay, Read 1 isolates the cell barcode and UMI, leaving only Read 2 to capture the actual transcript sequence ( $L$ ).

Traditional alignment algorithms<sup>7</sup> (e.g., BWA [23], Bowtie2 [24]) map these sequences base-by-base to assign exact genomic coordinates, typically relying on a Burrows-Wheeler Transform<sup>8</sup> index. While building this index is a one-time operation scaling as  $\mathcal{O}(T \cdot L_{avg})$ , querying it is computationally expensive. Because the algorithm must accommodate sequencing errors and mutations, it uses a branch-and-bound search that scales logarithmically with the reference size. Consequently, the total query time becomes:

$$\mathcal{O}(T \cdot L_{avg}) + \mathcal{O}(R \cdot L \log T)$$

For the billions of reads generated in a single-cell experiment, this error-tolerant sequence searching becomes a severe bottleneck, making exhaustive alignment prohibitively slow.

Modern transcriptomics tools like Salmon, alevin-fry, and kallisto [25, 26, 5] bypass this bottleneck through pseudo-alignment. This approach relies on a key insight: downstream single-cell quantification only requires identifying the source transcript, not its exact genomic coordinates. By discarding base-level alignment, pseudo-alignment replaces logarithmic sequence searches with constant-time hash table lookups.

This approach operates in two computational phases. First, index construction decomposes the reference transcriptome into a colored de Bruijn graph<sup>9</sup> that maps short sequences of length  $k$  ( $k$ -mers) to their parent transcripts. Rather than materialising every theoretically possible sequence, the index only stores the  $k$ -mers actually observed in the reference. This maintains a manageable spatial footprint of  $\mathcal{O}(d)$  for  $d$  distinct  $k$ -mers; for a mammalian transcriptome  $d \sim 10^8$ , far below the  $4^{31}$  theoretically possible  $k$ -mers. Second, during the query phase, each read is split into overlapping  $k$ -mers (roughly 60 to 120 per read). The algorithm queries the hash table for each  $k$ -mer and

<sup>7</sup> **Read alignment:** Base-by-base mapping of a read to the reference, giving exact coordinates at high computational cost.

<sup>8</sup> **Burrows-Wheeler Transform (BWT):** A reversible text transform that allows fast substring search in a compressed reference index.

<sup>9</sup> **Colored de Bruijn graph:** A graph of overlapping  $k$ -mers in which each  $k$ -mer is coloured by the transcripts containing it, used as the pseudo-alignment index.

intersects the results to rapidly identify an equivalence class<sup>10</sup>—the set of reference transcripts compatible with that read.

Because this method shifts the core operation from sequence alignment to hash retrieval, the per-read query cost decouples entirely from the size of the reference transcriptome. The total execution time simplifies to:

$$\mathcal{O}(T \cdot L_{avg}) + \mathcal{O}(R \cdot L)$$

The practical impact of this reduction is substantial. Using the baseline scale of  $R \sim 10^9$  reads,  $L \sim 10^2$  bp, and  $T \sim 10^5$  (where  $\log_2 T \approx 16.6$ ), traditional alignment’s query phase requires approximately  $1.7 \times 10^{12}$  operations. In contrast, pseudo-alignment requires roughly  $1.0 \times 10^{11}$  operations. This 50- to 100-fold performance leap occurs because pseudo-alignment drops the asymptotic  $\log_2 T$  factor and replaces expensive string matching with efficient hash lookups, ultimately reducing processing times from days to hours.

### Phase 2.1: Probabilistic Resolution via Expectation-Maximization

Pseudo-alignment introduces a distinct computational challenge: multi-mapping reads<sup>11</sup>. These occur when a read’s  $k$ -mers match multiple transcripts that share exonic sequences. If a read falls entirely within a shared exon, it carries no sequence information to identify its specific isoform of origin, prompting the pseudo-aligner to return a set of valid candidates rather than a single definitive transcript. This ambiguity is particularly severe in 3’-biased protocols, where short reads exclusively capture the terminal region and frequently match every isoform sharing that terminal exon [1].

Discarding these ambiguous reads is statistically unsound. Because they constitute the vast majority of alignments in 3’-biased data, removing them would severely deplete the count matrix and artificially bias isoform proportions toward rare variants that happen to carry unique sequences. Instead, modern pseudo-aligners resolve this ambiguity by framing quantification as maximum-likelihood estimation over a generative model of the sequencing experiment [25].

In this model, each read fragment is assumed to be produced in two stages. First, a transcript  $t$  is drawn with probability proportional to its abundance  $\theta_t$ . Second, a start position is drawn uniformly from the  $\ell_t$  valid positions where a fragment of the observed length could begin. This admissible count,  $\ell_t$ , represents the transcript’s<sup>12</sup>. Crucially, the true transcript of origin remains latent; the pseudo-aligner only reports the equivalence class  $C_r$  [26]. Marginalising over the unobserved origin yields the probability of observing that class under a given abundance vector:

$$P(C_r | \theta) = \frac{\sum_{t \in C_r} \theta_t \ell_t}{\sum_{t'} \theta_{t'} \ell_{t'}}$$

Because the origin of each read is unobserved, this likelihood is optimised using the Expectation-Maximization (EM)<sup>13</sup> algorithm, which alternates between two phases until convergence. In the

<sup>10</sup> **Equivalence class:** The set of transcripts that could have produced a read, since all contain the read’s  $k$ -mers.

<sup>11</sup> **Multi-mapping read:** A read compatible with more than one transcript because those transcripts share its sequence.

<sup>12</sup> **Effective length:** The number of fragment start positions a transcript can generate; approximately its length minus the fragment length, plus one. It corrects for the fact that longer transcripts yield more fragments at equal abundance.

<sup>13</sup> **Expectation-Maximization (EM):** An iterative algorithm that estimates transcript abundances and resolves ambiguous read assignments jointly until they converge.

Expectation step (E-step)<sup>14</sup>, the current abundance estimate is used to compute the posterior probability that read  $r$  originated from transcript  $t$ , given that it belongs to the class  $C_r$ :

$$\hat{z}_{rt} = \frac{\theta_t \ell_t}{\sum_{t' \in C_r} \theta_{t'} \ell_{t'}} \quad \text{for } t \in C_r,$$

and  $\hat{z}_{rt} = 0$  otherwise. The effective-length term  $\ell_t$  is critical here: it accounts for the fact that a longer transcript contributes proportionally more fragments at equal abundance, ensuring  $\hat{\theta}$  is not biased toward shorter isoforms. The algorithm then proceeds to the Maximization step (M-step), where each abundance is re-estimated as the total posterior weight assigned to its transcript:

$$\theta_t \propto \sum_{r: t \in C_r} \hat{z}_{rt}$$

These values are subsequently normalised so that  $\theta_t$  sums to exactly one across the transcriptome.

Two properties of this iteration influence the interpretation of the resulting counts. First, because each EM round is guaranteed not to decrease the log-likelihood, the estimates converge monotonically to a local optimum, though not necessarily a global one [27]. Since the likelihood surface is frequently flat or multimodal, implementations typically initialise  $\theta$  uniformly and stop when the relative change in log-likelihood falls below a specified tolerance. Second, the model relies on a closed-world assumption: every read is assumed to originate from an annotated transcript. Consequently, incomplete references or unannotated isoforms propagate directly into  $\hat{\theta}$ . The resulting point estimate lacks intrinsic uncertainty, meaning downstream methods that require variance—such as the differential transcript usage (DTU) tools discussed later—must recover it by bootstrapping the quantification step [28].

Computationally, because the EM updates run over equivalence classes rather than individual reads, the algorithm scales with the number of distinct classes. Tools like *alevin-fry*'s parsimonious variant exploit this by collapsing transcripts that share identical  $k$ -mer sets into single groups, effectively performing gene-level quantification where isoform resolution is impossible [5].

This collapsing behaviour highlights a fundamental identifiability limit inherent to 3' capture protocols. Consider two isoforms,  $t$  and  $t'$ , of the same gene whose reads fall entirely into equivalence classes containing both—a common scenario when a shared terminal region covers their effective lengths. During the E-step, posterior weight is assigned solely through the ratio  $\theta_t \ell_t / (\theta_t \ell_t + \theta_{t'} \ell_{t'})$ . If  $\ell_t \approx \ell_{t'}$ , this ratio reduces to  $\theta_t / (\theta_t + \theta_{t'})$ . As a result, the likelihood depends on the two abundances only through their sum and remains flat along  $\theta_t + \theta_{t'} = \text{constant}$ . Every proportional split of abundances explains the observed reads equally well. This identifiability problem is bounded by the sequence divergence at the terminal region, not by the estimator, the reference, or sequencing depth; sequencing the same fragment more deeply cannot separate two transcripts that the fragment itself cannot distinguish. Consequently, transcript-level analyses of 3'-biased data can only reliably recover isoform switches that differ at or near their 3' ends.

<sup>14</sup> **E-step / M-step:** The two alternating EM phases: distribute reads by the current abundance estimate (E), then re-estimate abundances (M).

## Phase 3: Deduplication and Matrix Generation

Once each read has been assigned to a transcript, the remaining task is to count molecules rather than reads. The protocol attaches a *UMI*—a short random sequence—to each captured molecule before amplification, so every read carries both a *cell barcode* identifying its cell and a UMI identifying its molecule. Because PCR copies inherit both tags, the reads descending from one molecule are indistinguishable duplicates, and collapsing each such group recovers the molecule counts that sequencing had inflated.

The collapse must be performed on barcode, UMI *and* assigned transcript together. A UMI holds only a few random bases, so within one cell two unrelated molecules can share a UMI by chance; grouping on barcode and UMI alone would merge them, whereas including the transcript keeps them apart whenever they originate from different features.

The result is the expression count matrix  $X$ , whose entry  $x_{fc}$  records the number of distinct molecules observed for feature  $f$  in cell  $c$ . Amplification no longer distorts the counts, so the differences remaining between cells reflect biology and capture efficiency rather than how often each molecule happened to be copied.

## COTAN Framework

The transcript-level matrix  $X$  produced above acts as the natural input for the statistical framework presented here, yet it simultaneously introduces the primary analytical difficulty: the shallow per-cell depth heavily imposed by the droplet hardware leaves the vast majority of its entries at zero. The COTAN framework, introduced by Galfrè et al. [1], was specifically engineered to analyse this zero-heavy structure directly, and has since been logically extended to recover entire cell clusters purely from co-expression patterns [29]. The following subsections formalise this sparsity and detail the contingency-table machinery COTAN employs to successfully bypass traditional density-based statistics.

### The Sparsity Problem

Single-cell count matrices exhibit extreme structural sparsity. The disparity between the reference transcriptome and the yield of a single cell sets the scale: if  $T \sim 10^5$  annotated isoforms sit in the reference while a single cell recovers only  $T_{obs} \sim 10^3$  distinct transcripts, raw alignment space carries an initial zero rate of approximately 99%. Standard preprocessing pipelines subsequently filter out unexpressed or ultra-rare features, condensing the effective feature space to  $T_{filtered} \approx 1.2 \times 10^4 - 2.3 \times 10^4$  viable transcripts, yet the observed zero rate typically remains between 70% and 90% [1].

These zeros fall into two categories: biological zeros<sup>15</sup> (where the cell did not express the transcript) and technical *technical dropouts* (where the molecule was present but failed to be captured). Because the cumulative efficiency of droplet capture, reverse transcription, and sequencing is only 10% to 20% [30], technical dropouts dominate. Once recorded, both types of zeros are numerically identical in the count matrix and cannot be distinguished by count-based tests alone: a transcript that was truly absent and one that was merely missed leave the same mark [31]. Furthermore, the 3' terminus<sup>16</sup>

<sup>15</sup> **Biological zero:** A zero count reflecting genuine absence of a transcript in a cell, as opposed to a technical dropout.

<sup>16</sup> **3' (three-prime) terminus** is the end of a transcript at which the poly-A tail is added; capture chemistry anchored here reads only the terminal fragment, not the transcript body.

*tag bias* exacerbates this sparsity by collapsing distinct isoforms that share terminal exons into a single observational unit. That ambiguity, rather than the sparsity itself, is the fundamental statistical barrier for conventional tools designed for deep, dense *bulk RNA-seq* data.

Applying standard density-based statistical methods to such sparse matrices introduces severe, compounding computational artifacts. Log-normalisation, which strictly involves adding a pseudo-count before logarithmic scaling, heavily distorts low-abundance measurements and artificially compresses legitimate zeros into spurious scaling patterns [1]. Standard linear or rank-based association measures, such as Pearson or Spearman correlations, fail entirely in this domain, as they inevitably end up measuring the coincidence of technical dropouts rather than identifying true regulatory co-expression [1]. Furthermore, attempting to artificially densify the matrix through heuristic imputation—specifically replacing zeros with smoothed estimates derived from neighbouring cells—frequently introduces fabricated correlations and severely compromises sensitive downstream signals [1]. To fundamentally circumvent these pitfalls, COTAN explicitly abandons imputation and variance-stabilising transformations entirely, choosing instead to evaluate the zeros as a direct signal through strict binary discretisation.

## Contingency Table Solution

As stated, COTAN cleanly bypasses the flawed continuous count amplitudes by evaluating pairwise feature dependencies using asymptotic independence testing applied strictly to binary presence and absence states [1]. For any given feature pair  $(f, f')$  across  $N$  cells, the continuous counts are logically reduced to discrete indicator variables for each cell  $c$ :

$$a_c = \mathbf{1}[x_{fc} > 0], \quad b_c = \mathbf{1}[x_{f'c} > 0]$$

where  $a_c$  marks whether feature  $f$  is detected in cell  $c$  and  $b_c$  whether its partner  $f'$  is. Letting the discrete states be indexed by  $i, j \in \{0, 1\}$ , the framework constructs a comprehensive  $2 \times 2$  *contingency table* where  $O_{ij}$  directly denotes the observed number of cells exhibiting state  $a_c = i$  and  $b_c = j$ . The marginal totals for this specific table are defined as  $\text{Row}_i = \sum_j O_{ij}$  and  $\text{Col}_j = \sum_i O_{ij}$ , mathematically constrained by the total cell count  $\sum_i \text{Row}_i = \sum_j \text{Col}_j = N$ . Under the standard null hypothesis of statistical independence, the expected cell count  $E_{ij}$  is calculated directly as:

$$E_{ij} = \frac{\text{Row}_i \cdot \text{Col}_j}{N}$$

Table 2.3 lays the four counts out explicitly. The two diagonal cells hold cells in which the features are detected together ( $O_{11}$ ) or neither is detected ( $O_{00}$ ); the off-diagonal cells hold cells defining the difference between them, in which exactly one of the two is present. Co-expression resides in the tendency of the count to fall on a diagonal rather than on the anti-diagonal, and the expected value  $E_{ij}$  above is what that count would be under the independence hypothesis.

**Table 2.3:** The  $2 \times 2$  contingency table built for a feature pair  $(f, f')$  across  $N$  cells. Entry  $O_{ij}$  counts cells with  $a_c = i$  and  $b_c = j$ ; the margins are the feature’s detection totals.

|                            | $f'$ detected ( $b_c = 1$ ) | $f'$ absent ( $b_c = 0$ ) | Row total      |
|----------------------------|-----------------------------|---------------------------|----------------|
| $f$ detected ( $a_c = 1$ ) | $O_{11}$                    | $O_{10}$                  | $\text{Row}_1$ |
| $f$ absent ( $a_c = 0$ )   | $O_{01}$                    | $O_{00}$                  | $\text{Row}_0$ |
| Column total               | $\text{Col}_1$              | $\text{Col}_0$            | $N$            |

Deviation from this independent expectation is evaluated using Pearson’s chi-squared statistic [1]:

$$\chi^2 = \sum_{i \in \{0,1\}} \sum_{j \in \{0,1\}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

The result is reported not as a statistic but as a single coefficient. COTAN defines the co-expression coefficient  $C_{f,f'}$  by rescaling the deviation so that it lies in  $(-1, 1)$  and, under the null hypothesis, satisfies  $N C_{f,f'}^2 \sim \chi^2(1)$  over the  $N$  cells of the table. The pairwise  $p$ -value, the quantity used for screening throughout this thesis, then follows directly:

$$p_{f,f'} = P(\chi^2(1) > N C_{f,f'}^2)$$

A positive coefficient means the two features are detected together more often than independence predicts; a negative one means less often, which is the mutual exclusivity that Chapter 3 treats as the signal of interest.

One modification separates  $C_{f,f'}$  from the textbook Phi coefficient of a  $2 \times 2$  table, and it lies in the expectation rather than in the statistic. Taking  $E_{ij}$  proportionally to the marginals, as above, assumes that every cell has the same chance of expressing a given feature. Cells differ widely in library size, so the cells that capture less RNA accumulate zeros across all features at once, and a proportional expectation would read that shared under-detection as co-expression and inflate the coefficient. COTAN therefore derives the expected co-occurrence from a model of the zeros themselves, calibrated per feature and per cell from the zeros actually observed; the reader is referred to the COTAN paper for the exact form of that model [1]. Co-expression is thus measured against how often two features’ zeros were expected to coincide, not against their raw marginals.

The chi-squared approximation behind  $p_{f,f'}$  is asymptotic in the number of cells, not in the counts, and the distinction matters here. With  $N$  in the thousands the calibration is reliable for features detected in a reasonable fraction of cells, which covers most of a standard gene matrix. It degrades in the opposite regime, where a feature is rare enough that several expected cell counts  $E_{ij}$  fall below about five; Pearson’s statistic is then anti-conservative, and such a pair is best read as inconclusive rather than significant. Filtering cannot remove this effect for the rarest transcripts.

## Output Metrics

The framework reduces the machinery above to three output metrics [1].

The co-expression matrix  $C$  captures the pairwise associations between features through the calculated coefficient  $C_{f,f'}$ . Within this matrix, positive values represent co-expression, whereas negative values indicate mutual exclusivity.

The *Global Differentiation Index (GDI)* is calculated from the co-expression matrix prior to any cellular clustering and functions as a per-feature measure of cluster specificity. For feature  $i$  it takes the mean of the smallest 5% of the feature’s pairwise  $p$ -values,  $\bar{p}_i$ , and applies the transformation  $\text{gdi}_i = \ln(-\ln \bar{p}_i)$  to compress the distribution into an interpretable scale [1]. Features with constitutive expression form a uniform lower tail around a null median of roughly 1.305. Conversely, features that vary across subpopulations produce very small  $p$ -values and rise above approximately 1.5, which separates cell-type markers from housekeeping features.

Differential expression analysis (DEA) determines whether a single feature  $f$  is statistically enriched within a specific cell cluster  $S$ . It applies the contingency-table structure to compare the presence or absence of feature  $f$  against cells located inside versus outside cluster  $S$ , yielding the correlation

coefficient  $D_{f,S}$  on the bounded scale  $(-1, 1)$ . A positive  $D_{f,S}$  reveals that the feature is detected more frequently inside the cluster, while a negative value signifies it is detected less often. Because all DEA coefficients share this standardized scale, the enrichment values can be directly compared across different features and clusters.

COTAN also provides functions for cell clustering, separate from these metrics but built on the GDI: a uniform-clustering step that groups cells into clusters whose gene expression is homogeneous, checked by ensuring no more than a small fraction of genes exceed a GDI threshold, followed by a merging step that collapses clusters which remain uniform when joined. DEA operates on whichever clusterization is supplied, so the two are connected but distinct capabilities.

## The Theoretical Basis of DTU in Sparse scRNA-seq

### The Probabilistic Definition of a Switch

Differential Gene Expression (*DGE*) and Differential Transcript Usage (*DTU*) target fundamentally different mathematical properties of the transcriptome. DGE strictly measures overall changes in total, aggregated gene abundance between conditions; if  $\theta_g$  represents the total absolute abundance of a parent gene  $g$ , DGE models shifts in  $\theta_g$  directly. DTU, conversely, models the conditional probability of observing a specific transcript  $t$  given that its parent gene is expressed. Let  $\theta_t$  represent the absolute abundance of transcript  $t$ ; the continuous relative proportion of that transcript is therefore  $P(t | g) = \theta_t / \theta_g$ . A theoretical isoform switch occurs when the conditional probability  $P(t_1 | g)$  of one variant significantly increases between biological conditions or cellular states while  $P(t_2 | g)$  simultaneously decreases, a regulatory shift that can occur entirely independently of any change in the total aggregated abundance  $\theta_g$ .

### Translating Continuous Proportions to Binary Sparsity

Traditional bulk RNA-seq tools model the continuous proportion  $P(t | g)$  directly, a statistical approach that relies on deep, junction-spanning coverage to provide stable density. In the extreme sparsity of droplet data (Section 2.2.1), a continuous proportional model breaks down. Because the cumulative efficiency of droplet capture and sequencing is exceedingly low, as an isoform's relative proportion drops, its already-low integer count is overwhelmingly pushed entirely into the zero state, registering as *technical dropout*. Consequently, a continuous regulatory shift in biological proportions manifests mathematically in the single-cell count matrix as asymmetric binary presence and absence across cells.

### The Theoretical Limit of Observability

The underlying pseudo-alignment quantification model imposes a strict theoretical boundary on which DTU events are mathematically identifiable from the matrix. As established in Section 2.1.3, the pseudo-aligner maps sequences not to exact coordinates but to equivalence classes  $C_r$ , representing the set of all transcripts compatible with a given read. If two actively switching isoforms  $t_1$  and  $t_2$  differ strictly in their internal exonic structure but share identical sequences at their 3' terminus, all terminal reads generated by droplet protocols will invariably fall into an equivalence class containing both variants. During quantification, the Expectation-Maximization (EM) algorithm assigns posterior weights based on the length-adjusted ratio  $\theta_{t_1} \ell_{t_1} / (\theta_{t_1} \ell_{t_1} + \theta_{t_2} \ell_{t_2})$ . If their effective lengths are approximately equal ( $\ell_{t_1} \approx \ell_{t_2}$ ), the likelihood surface remains entirely flat along the constraint

$\theta_{t_1} + \theta_{t_2} = \text{constant}$ . Every proportional split of abundances explains the observed terminal reads equally well, rendering the true  $P(t \mid g)$  unidentifiable. Therefore, the framework is theoretically constrained: it can only successfully detect DTU events—such as alternative 3' terminus usage or terminal exon variation—where sufficient sequence divergence allows the short terminal fragments to map to distinct, informative equivalence classes.

## Mutual Exclusivity Signal

Within the adapted COTAN framework, negative statistical associations—formally defined as *mutual exclusivity*—between features serve as the foundational indicator of alternative usage patterns. When applied specifically to multiple isoforms of the exact same parent gene, this structural mutual exclusivity is biologically consistent with alternative splicing or *isoform switching* events, wherein different cellular states preferentially express entirely distinct transcript variants.

However, mutual exclusivity in isolation is insufficient to prove a functional switch, since a strong negative association can also arise from sparse detection rather than from a real switch. One way to demonstrate the switch, consistent with the  $P(t \mid g)$  definition, is bidirectional opposition: a cluster in which one isoform is preferred and another in which the preference flips to its partner. This confirms that different cell types genuinely use different isoforms, rather than leaving the exclusivity explainable by sparsity.

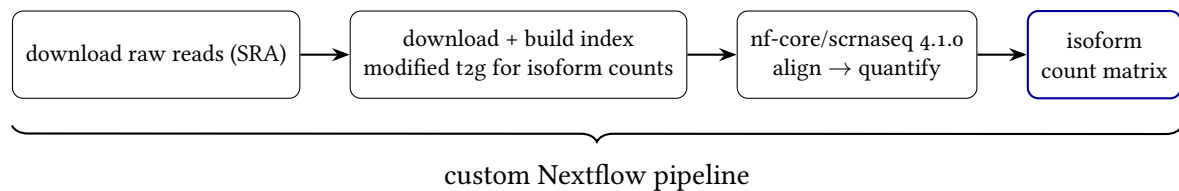


## Chapter 3

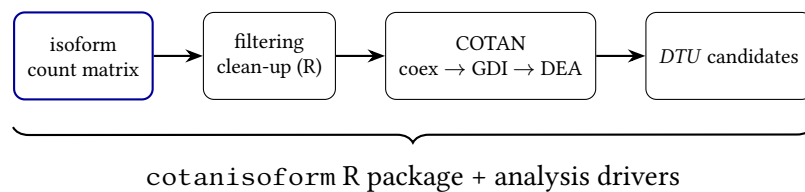
# Materials and Methods

Before walking through the workflow stages one by one, an overview of the whole is worth a glance: Figures 3.1 and 3.2 split it into two halves by the one artifact they share—an unfiltered isoform count matrix. The first half turns raw *FASTQ* reads into transcript-level counts (Section 3.2); the second takes that matrix, filters and cleans it, and runs the COTAN analysis and turns the counts into *DTU* candidates (Sections 3.3 and 3.5 onward).

Each half is implemented by a separate piece of code. The first half is driven by a custom Nextflow pipeline written for this thesis: it chains download, index, alignment and quantification around `nf-core/scrnaseq` (v4.1.0), runs on the cluster, and ends where the raw count matrix is written. The second half is driven by the `cotanisoform` R package together with config-driven analysis scripts, both written for this thesis; they consume that matrix, filter and clean it, and run COTAN to yield the *DTU* candidates. The two halves therefore never share a runtime: the heavy sequencing work lives in Nextflow, the statistical work in R.



**Figure 3.1:** Overview of the first half of the workflow: a custom Nextflow pipeline turns raw *FASTQ* reads into an unfiltered isoform count matrix.



**Figure 3.2:** Overview of the second half of the workflow: the isoform count matrix is filtered, cleaned, and processed through COTAN’s co-expression, GDI and DEA stages to yield the *DTU* candidates.

**Table 3.1:** Tools used in this thesis: role, version, and the section justifying each choice.

| Tool             | Version                      | Role                                 | Justified in |
|------------------|------------------------------|--------------------------------------|--------------|
| Nextflow         | 26.04.4                      | Pipeline orchestration               | 3.2          |
| nf-core/scrnaseq | 4.1.0                        | Automates index → align → quantify   | 3.2          |
| simpleaf         | 0.24.1 quant. / 0.24.0 index | Orchestrates alevin-fry + piscem     | 3.2          |
| alevin-fry       | —                            | Quantification; EM assignment        | 3.2          |
| piscem           | —                            | Index construction                   | 3.2          |
| sra-tools        | —                            | FASTQ download                       | 3.2          |
| Ensembl          | release 102                  | Reference annotation                 | 3.2          |
| R                | 4.5.3                        | Analysis layer                       | 3.3          |
| COTAN            | 2.13.1 (be93aa8)             | Contingency, co-expression, DEA, DTU | 3.3          |
| Seurat           | 5.5.1                        | Count-object container               | 3.3          |

## Data Acquisition and Dataset Selection

This study is based on two publicly available, droplet-based single-cell RNA-sequencing datasets: the Arigoni human cell-line mixture (GSE243665[32]) and the Ding mouse brain dataset (GSE132044[22]). Both were obtained as raw *FASTQ* reads alongside their genome annotations. They were selected after evaluating potential candidates against specific inclusion criteria, which are detailed below.

Candidate datasets were drawn from the curated collection at <https://seriph78.github.io/COTAN.Datasets.analysis/>, rather than searching the Gene Expression Omnibus (GEO)<sup>1</sup>[33] ad hoc. Importantly, these datasets are pre-validated. Their prior use in COTAN’s original benchmarking studies confirms their compatibility with the framework and ensures they are free of major technical artifacts.

A limitation of this collection is the absence of publicly available transcript-level count matrices for *scRNA-seq* data. This reflects standard field practices rather than an omission. By default, most pipelines—including CellRanger<sup>2</sup>[16] and the standard Seurat workflow[34]—collapse data to *gene*-level counts. While transcript quantification is possible using pseudo-alignment tools like Salmon[25] or alevin-fry[5], these steps are frequently bypassed. Furthermore, even when transcript-level analysis is performed, the data is typically aggregated back to the gene level prior to publication. Consequently, transcript matrices must be generated directly from the raw *FASTQ* files.

That rebuild can be automated from scratch with the nf-core/scrnaseq pipeline[35], a containerised Nextflow workflow that chains indexing, alignment and quantification behind a single invocation.

That choice, however, narrows the field: nf-core/scrnaseq supports barcode-based protocols only, with the 10x Genomics[16] chemistries, versions 2 and 3, as the supported family, so any dataset

<sup>1</sup> **Gene Expression Omnibus (GEO):** NCBI’s public database of published gene-expression datasets. It stores raw and processed data for re-analysis.

<sup>2</sup> **CellRanger / Seurat:** Standard pipelines (10x CellRanger, Seurat) that quantify and analyse *scRNA-seq* data, defaulting to gene-level counts.

analysed here must come from that chemistry family. Of the COTAN datasets, only the Arigoni human cell-line mixture and the Ding mouse brain dataset satisfy both this chemistry requirement and the availability of published barcode lists — the two datasets introduced above.

Those two are, in fact, a fortunate pairing. The Arigoni mixture is the strong-signal case: its cell lines are well separated and each carries a distinct isoform programme, and its GEO deposit supplies published barcode labels per line, so the detector runs against known groups and its calls can be checked against cell-type identity. The Ding cortex is the weak-signal case: heterogeneous tissue with gradual transitions and subtler contrasts, where no published labels exist, so it forces the computational-partitioning route and the cross-resolution validation. The pair therefore brackets the difficulty spectrum—a clean, discrete test and a noisy, continuous one—so that the candidates converging across both (Chapter 4) argue for a method that generalises rather than one fitted to a single favourable context.

## Raw Read Download

The raw input is not uniformly FASTQ. SRA stores 10x single-read runs as aligned BAM files rather than as reads, so the download path depends on each accession’s layout. A layout check picks the route: paired-end runs are fetched as FASTQs via `prefetch + fasterq-dump`, whereas single-end 10x runs are fetched as the TenX BAM (`prefetch -type TenX`) and converted back to FASTQs with `bamtofastq`[36]. The Arigoni and Ding deposits are both 10x single-end runs, so both take the BAM-then-`bamtofastq` path.

Two operational choices keep the downloads tractable. They stage in RAM (`/dev/shm`) and never touch persistent disk, because each sample is tens of gigabytes—about 18 GB of BAM or 8 GB of zipped FASTQ. And concurrency is capped to stay within SRA’s rate limits.

## Alignment, Quantification, and Feature Generation

### Algorithm Selection: `alevin-fry` (`simpleaf`) vs. `kallisto`

Quantifying transcripts from raw reads requires a choice between alignment-based and pseudo-alignment-based pipelines, and that choice propagates into the UMI handling and ambiguity resolution that downstream analysis depends on. Pseudo-alignment was preferred here for the reasons set out in Section 2.1.3. Two pseudo-aligners dominate single-cell transcript quantification, each pairing a pseudo-alignment engine with a UMI-deduplication step: `kallisto` with `bustools`, and `alevin-fry` with its `simpleaf` orchestrator. Table 3.2 compares their primary characteristics.

**Table 3.2:** Comparison of scRNA-seq pseudo-alignment architectures.

| Feature                | <code>kallisto</code> — <code>bustools</code> | <code>alevin-fry</code> ( <code>simpleaf</code> ) |
|------------------------|-----------------------------------------------|---------------------------------------------------|
| Multi-mapping handling | Full compatible set                           | Minimal set (parsimony)                           |
| UMI deduplication      | Standard collapse                             | Directional error modelling                       |
| Execution speed        | Fast (~10 min per sample)                     | Very fast (~5 min per sample)                     |
| nf-core integration    | Available, non-standard                       | Native (v4.1.0)                                   |

The `simpleaf` framework[37] (vo.24.1 for quantification, vo.24.0 for the index build), which orchestrates `alevin-fry` for quantification and `piscem` for indexing, was selected for three reasons. Its

UMI handling is superior, since single-cell data relies on *UMIs* to correct *PCR* amplification bias and the alevin-fry error model distinguishes true *UMIs* from sequencing errors more reliably. For reads compatible with multiple transcripts, alevin-fry resolves the ambiguity by parsimony, searching for the minimal set of transcripts that explains each UMI rather than counting it against every compatible transcript, which suppresses spurious low-abundance counts that would bias the within-gene proportions that *DTU* detection reads out. Finally, *3nf-core/scrnaseq* (v4.1.0) supports *simpleaf* natively, with fewer configuration quirks than the kallisto alternatives.

The parsimony resolution matters. Assigning a UMI to the full set of transcripts it is compatible with leaks counts from an abundant isoform onto lowly-expressed isoforms sharing its sequence, biasing the within-gene proportions that *DTU* detection reads out. *parsimony*<sup>4</sup> instead builds a parsimony graph and finds the minimal set of transcripts that explains each UMI, so spurious low-abundance counts are suppressed and the estimated proportions are better identified. The resolution can leave fractional counts on ties, which are rounded to integers before COTAN ingests the matrix.

Because the downstream analysis operates on binary presence and absence rather than on the abundance point estimates, bootstrapped abundance variance was not part of this pipeline and is of limited relevance to zero-based detection: the rounded counts are discretised into detection indicators before the co-expression statistics are computed.

## Index Modification for Transcript-Level Resolution

Standard alevin-fry[5] output collapses to gene level by default via transcript-to-gene mapping<sup>5</sup>. To extract transcript counts, the index files were modified before alignment, following the identity-mapping route discussed in the *simpleaf* issue tracker<sup>6</sup>:

1. **Identity mapping in `t2g.3col.tsv`:** the standard mapping is replaced by a self-referential one, which prevents the final aggregation step that sums isoforms into genes.

```
# Original: ENSMUST00000193814  ENSMUSG00000067548  # parent gene
# Modified: ENSMUST00000193814  ENSMUST00000193814  # itself
```

2. **Removal of `gene_id_to_name.tsv`:** the *nf-core* pipeline uses this file to map Ensembl gene identifiers to gene names during annotation. With the identity mapping in place the matrix rows are Ensembl *transcript* identifiers, so the annotation step finds no gene IDs to match and halts. Removing `gene_id_to_name.tsv` from the index trips a safety guard in alevin-fry: the annotation step checks for the file's presence and, finding it absent, skips the step. The guard does not itself know that annotation would halt on transcript IDs; we do, and we remove the file precisely to prevent that halt. The transcript counts were already written to disk, so they are kept and carried into the analysis.

This modification is **reversible**: transcript-level quantification is a finer partition of the gene-level signal, so summing the transcript rows by gene using the retained `t2g` and `gene_id_to_name` tables recovers a gene-level matrix directly, without rebuilding the index or re-aligning. This transcript-level route is an unsupported workaround, since alevin-fry is built for gene-level collapse and the identity-mapping modification is not an official mode. The gene-level matrix recovered by summing

<sup>3</sup> **nf-core:** A community repository of standardised, peer-reviewed Nextflow analysis pipelines.

<sup>4</sup> **parsimony:** An alevin-fry *UMI*-resolution mode that searches for the minimal set of transcripts explaining each UMI, suppressing spurious low-abundance counts.

<sup>5</sup> **Transcript-to-gene (t2g) mapping:** A file pairing each transcript with its parent gene; replacing it with an identity mapping prevents gene-level collapse.

<sup>6</sup> <https://github.com/COMBINE-lab/simpleaf/issues/115> (accessed 29 September 2026)

the transcript rows therefore matches a native gene-level run, except where the two resolutions assign ambiguous reads differently.

## Reproducible Workflow Automation

To make this workflow reproducible across datasets, it was automated with **Nextflow**[38] (v26.04.4), a domain-specific language for computational pipelines that handles parallelization and checkpointing. This choice provides error handling, containerized dependencies, and standard configuration across datasets.

The custom automation script:

- **Input:** a CSV samplesheet listing each sample and its SRA accessions<sup>7</sup>, and a per-dataset configuration naming the genome assembly and species, the Ensembl release (102), the 10x chemistry, and the published barcode list. The pipeline downloads the reference FASTA and GTF<sup>8</sup> from Ensembl itself, so a matching assembly name replaces a manual reference download. The configuration is not limited to these fields: it also carries options for the alignment mode, the index strategy, and other pipeline parameters, which are left at sensible defaults unless a dataset needs them.
- **Automated steps:**
  1. Download the raw reads (Section 3.1): FASTQs for paired-end runs, BAM-then-bamtofastq for 10x single-end runs.
  2. Download the Ensembl reference and build the gene-level `piscem` index.
  3. If transcript-level output is requested, create a cheated transcript-level copy of that index by applying the identity-mapping modification.
  4. Execute `nf-core/scrnaseq` (v4.1.0) with `simpleaf`.
  5. Produce the raw count matrix and its metadata.
- **Availability:** the complete workflow, including configuration files and test cases, is available at <https://github.com/ldelitala/sc-isoform-cotan>.

The workflow ends at the count matrix. Quality filtering and the COTAN analysis are performed afterwards in the R analysis layer (R 4.5.3), keeping the two concerns separate.

Adapter trimming is omitted deliberately rather than by oversight. The 10x read structure places the barcode and UMI in a fixed header segment, and `simpleaf` resolves that geometry internally with a UMI-aware error model, discarding residual adapter  $k$ -mers because they match no indexed sequence. A separate `fastp` pass would therefore reread the entire FASTQ file for no change in the count matrix. Quality control at the barcode level still runs: `simpleaf` applies empty-droplet detection to the quantification output, and cells are further restricted to the high-quality barcode sets published with each dataset (Section 3.3).

<sup>7</sup> **Sequence Read Archive (SRA):** NCBI's repository of raw sequencing data, accessed by identifiers such as SRR accessions.

<sup>8</sup> **GTF (Gene Transfer Format):** An annotation file listing each gene's coordinates, exon structure, and transcript boundaries; used to build the transcriptome index.

## Data Processing and Quality Control

Before the COTAN object is built, the Seurat assay's counts are rounded to the nearest integer and explicit zeros are dropped, since COTAN ingests integer counts; the fractional parsimony-em assignments never enter the framework directly. The COTAN object is then initialised from this Seurat object, and all subsequent filtering operates on it. From this point onward Seurat is no longer used; the rest of the workflow operates exclusively on the COTAN object.

### Cell-Level Filtering

Cells were filtered using metadata from the original GEO publications. The Arigoni dataset (GSE243665[32]) and the Ding dataset (GSE132044[22]) both provide lists of cells that passed their quality control thresholds, and restricting the analysis to those known-good barcodes avoids a second QC pipeline that could introduce inconsistencies. Published studies typically filter on a minimum UMI count per cell, which removes empty droplets; a maximum mitochondrial read percentage<sup>9</sup>, which removes dying cells; and a gene detection threshold, since cells with too few detected genes may be doublets or otherwise low quality. Reusing their filtering ensures comparability and reduces methodological drift.

The unfiltered matrices are dominated by empty droplets; restricting to the barcodes the original studies report as passing their quality control yields the cell counts. Concretely, the passing-cell barcode lists were loaded into the COTAN object as a `passed_QC` condition using the package's `flag_valid_cells_from_tsv()`, and the object was then subset with `filter_cotan_by_condition()` to keep only cells flagged as passing; the per-cell-line Arigoni lists were combined first. Both functions are part of the `cotanisoform` R package[39], built on top of COTAN's object model.

### Feature-Level Filtering

Feature filtering ran in the R analysis layer (R 4.5.3) with COTAN (v2.13.1, commit be93aa8). Two filters act on the transcript set. First, `filter_spliced_transcripts()` keeps only spliced transcripts by matching the Ensembl feature names against a prefix (ENST for human, ENSMUST for mouse) and discarding the rest, then drops the cells that are left with no surviving feature. COTAN's framework[1] targets mature mRNA, and unspliced pre-mRNA has different statistical properties that can confound *co-expression* analysis, so this restriction to spliced features both removes noise and shrinks the matrix. Second, `clean_cotan_data()` wraps COTAN's `clean()` routine with its default cutoffs: a transcript is kept only if it is expressed in at least 0.3% of cells (`cellsCutoff=0.003`) and a cell only if it detects at least 0.2% of the retained features (`genesCutoff=0.002`)—fractional cutoffs far stricter than a fixed count, because isoform-level detection is so sparse that an absolute threshold would admit thousands of near-empty rows. With `drop_fully_expressed` enabled, the few transcripts expressed in nearly every cell are physically removed as well.

These choices prioritize quality over quantity. Smaller matrices mean faster computation and cleaner signal—important when running expensive contingency table calculations across all feature pairs.

<sup>9</sup> **Mitochondrial read percentage:** The fraction of a cell's reads coming from mitochondrial genes; abnormally high values flag dying or damaged cells.

## Statistical Estimation and the GDI

With the filtered transcript matrix in the COTAN object, the framework’s parameters are estimated and its statistics computed. The per-feature and per-cell expression probabilities  $\lambda$  and  $\nu$  are calibrated from the observed zeros, the pairwise co-expression coefficient  $C_{f,f'}$  and its  $p$ -value are calculated for every feature pair, and the *GDI* is derived from those  $p$ -values as the per-feature mean of the smallest 5%—the theory is laid out in Section 2.2. The pipeline executes all of this in one step (`03_calc.R`) and renders the resulting GDI into a diagnostic plot (`04_gdi.R`) over a dataset-specific marker panel. This is the stage that turns the filtered counts into the co-expression structure the partition in the next section reads out of; without it, clustering would have nothing to group cells by.

## Population Partitioning and Validation

Differential transcript usage is a contrast between groups of cells: an isoform switch surfaces only when the cells carrying one variant can be set against the cells carrying the other. The partition of cells fixes what the detector is able to see, and the two datasets reach that partition along different routes.

### Ground-Truth vs. Computational Partitioning

The Arigoni human cell-line dataset arrives already divided: its GEO deposit distributes one barcode list per constituent cell line and for the PBMCs, so those published labels serve directly as the groups the detector compares.

The Ding mouse cortex dataset arrives with no such labels. Its cells are grouped from the matrix itself with COTAN’s hierarchical uniform clustering<sup>10</sup>. Cell-to-cell dissimilarity is read from the co-expression patterns, clusters are split until each becomes homogeneous, and neighbours the statistics cannot separate are merged. Run on transcript counts, the procedure returns a set of uniform clusters and a small noisy remainder—the cells whose expression matches no group closely enough to claim them. Those clusters are a hypothesis, not a verdict: COTAN was designed and validated on gene-level matrices, and this thesis is its first application at isoform level, so nothing guarantees the co-expression statistics still hold on transcripts.

### Cross-Resolution Validation

To give the transcript partition something to be measured against, the same cells were clustered a second time on gene counts: the official Ding gene matrix, cleaned with COTAN’s standard `clean()` and clustered with parameters identical to the transcript run. Gene-level clustering is the established use of COTAN; over a denser feature set it stands as an independent reference partition rather than a cheaper substitute. The two partitions are then compared over their common cells by a confusion matrix<sup>11</sup>, and the detector is run twice over the same transcript matrix, once with the transcript clusters and once with the gene-derived clusters standing in as ground truth, so that an event surviving both is a signal no choice of partition can manufacture.

<sup>10</sup> **Hierarchical uniform clustering:** COTAN’s procedure that recursively splits cells by co-expression dissimilarity, then merges statistically indistinguishable clusters.

<sup>11</sup> **Confusion matrix:** A table of cluster-label agreement between two clustering runs, revealing systematic splitting and merging.

## The DTU Detection Algorithm

The detector bypasses pseudo-bulk aggregation and instead reuses COTAN’s gene-level machinery—the contingency table, the *co-expression* coefficient, and the differential-expression contrast—applied at transcript resolution, so that a mutually exclusive isoform pair is recognised by the same independence test that already copes with sparsity[1]. The biological justification for the two filters is given in Section 2.3.4. For every gene with  $k \geq 2$  detected isoforms, the procedure is:

1. **mutual exclusivity screening:** for every possible pair of isoforms  $(A, B)$  within the gene:
  - Read the *co-expression* coefficient  $C_{A,B}$  already computed for every feature pair in the estimation stage (Section 3.3), where it derives from the  $2 \times 2$  contingency table of presence and absence across cells[1] (Section 2.2).
  - Retain pairs with a negative  $C_{A,B}$  and a  $p$ -value<sup>12</sup>  $< 0.05$ .
2. **Contrast calculation:** for each significant pair, the differential enrichment is a vector over the partition’s clusters,

$$\Delta_i = D_{A,i} - D_{B,i}, \quad i \in \mathcal{P}, \quad (3.1)$$

where  $D_{f,i}$  is COTAN’s differential expression coefficient for feature  $f$  in cluster  $i$  (Section 2.2) and  $\mathcal{P}$  is the set of clusters in the partition. Each term  $D_{f,i}$  lies in  $(-1, 1)$  because it is a correlation coefficient read from the  $2 \times 2$  contingency table of feature presence against cluster membership, and such a coefficient is bounded to  $(-1, 1)$  by construction; the difference  $\Delta_i$  therefore lies in  $(-2, 2)$ , so the contrast is bounded and comparable across genes and clusters.

3. **Bidirectional thresholding:** retain pairs that switch, which requires opposition in both directions:
  - At least one cluster with  $\Delta > \tau$ .
  - At least one different cluster with  $\Delta < -\tau$ .
  - <sup>13</sup>  $\tau$  is calibrated per dataset in the section that follows.

The algorithm outputs a list of gene-isoform pairs with evidence for cluster-specific differential usage, ranked by statistical significance and effect size ( $|\Delta|$ ).

## Calibrating the Switch Threshold ( $\tau$ )

The contrast threshold  $\tau$  controls the balance between sensitivity and specificity: high values yield few confident calls but risk missing subtle switches; low values capture more events but increase false positives.

A tuning script was developed to study how the threshold affects the candidate set. It varies  $\tau$  in fine steps across a range such as 0.02 to 0.5 (step 0.02 for the Arigoni run, 0.01 over 0.01–0.3 for the Ding run), counts the detected DTU events at each threshold, plots the relationship between  $\tau$  and the number of significant pairs, and suggests thresholds that yield biologically reasonable candidate counts, typically 20–100 events for validation feasibility.

<sup>12</sup> **p-value:** The probability of observing an association this strong if the features were truly independent.

<sup>13</sup>  **$\tau$  (contrast threshold):** The minimum absolute DEA contrast between two isoforms required to count a cluster toward a switch.

## Chapter 4

# Results and Discussion

This thesis shows that sparsity-robust, contingency-table statistics yield positive results for recovering *DTU* from 3'-biased single-cell RNA-seq data. Applied to both a human cell line mixture and mouse cortical tissue, the adapted framework identified biologically consistent, mutually exclusive isoform switches, including *TPM1*, *Meg3*, and *Malat1*.

However, the analysis does not jump straight to these core findings. Because the dataset relies on highly sparse terminal counts, every DTU call is only as trustworthy as the quality and clustering checks beneath it. Therefore, this chapter is structured as a fixed sequence of checks, with each phase guarding the step that follows. It first states the dimensions of the filtered matrices that bound every downstream step (Section 4.1), then evaluates the *GDI* to confirm that the raw transcript counts carry enough differential signal to separate distinct cell populations (Section 4.2). It then compares clustering partitions across gene and transcript matrices to locate exactly where isoform information alters the cellular grouping (Section 4.3). Only after these foundational tests validate the matrix's integrity does the analysis proceed to extract the DTU candidates (Section 4.4) and stress-test them against contrast thresholds (Section 4.4) and the published literature (Section 4.5).

### Filtered Matrix Dimensions

The pipeline's output is fixed by the two filtering passes whose sizes bound everything downstream, and this section walks through them in the same order as Section 3.3 describes them. Tables 4.1 and 4.2 collect the sizes at each step, one per dataset, so each number can be traced to the pass that produced it.

| Step                          | Barcodes | Transcripts |
|-------------------------------|----------|-------------|
| Raw matrix                    | 488,342  | —           |
| Cell-level pass (passing QC)  | 29,606   | —           |
| Feature-level pass (filtered) | 29,454   | 23,673      |

**Table 4.1:** Arigoni matrix dimensions per filtering step, in the order of Section 3.3. The raw transcript count is not part of the reported set; the feature-level pass is the final matrix.

Although the transcriptome lists more transcripts than genes, the cleaned transcript matrix ends up smaller than the gene matrix: the raw mouse matrix detects 149,443 transcripts, yet after filtering only 12,848 remain, against 28,692 genes in the authors' GEO count set—isoform-level detection is so much sparser that most transcripts never pass the sparsity cutoff.

| Step                          | Barcodes | Transcripts |
|-------------------------------|----------|-------------|
| Raw matrix                    | 121,082  | 149,443     |
| Cell-level pass (passing QC)  | 4,091    | —           |
| Feature-level pass (filtered) | 4,077    | 12,848      |

**Table 4.2:** Ding matrix dimensions per filtering step, in the order of Section 3.3; 28,692 genes in the authors’ GEO set are shown in the text for the gene-level contrast.

## Dataset Quality Assessment via GDI

Before DTU calls can be interpreted, the matrix itself must be shown to carry enough variability to distinguish cell populations. The *GDI*, defined in Section 2.2, provides this check without any clustering: it measures how much each feature varies across cells, so constitutive features sit near the null median (about 1.305) while features that mark distinct populations rise above the threshold. The question here is whether the transcript-level counts still separate the populations once the gene-level resolution is replaced by sparse isoform counts.

### Transcript-level GDI Distribution

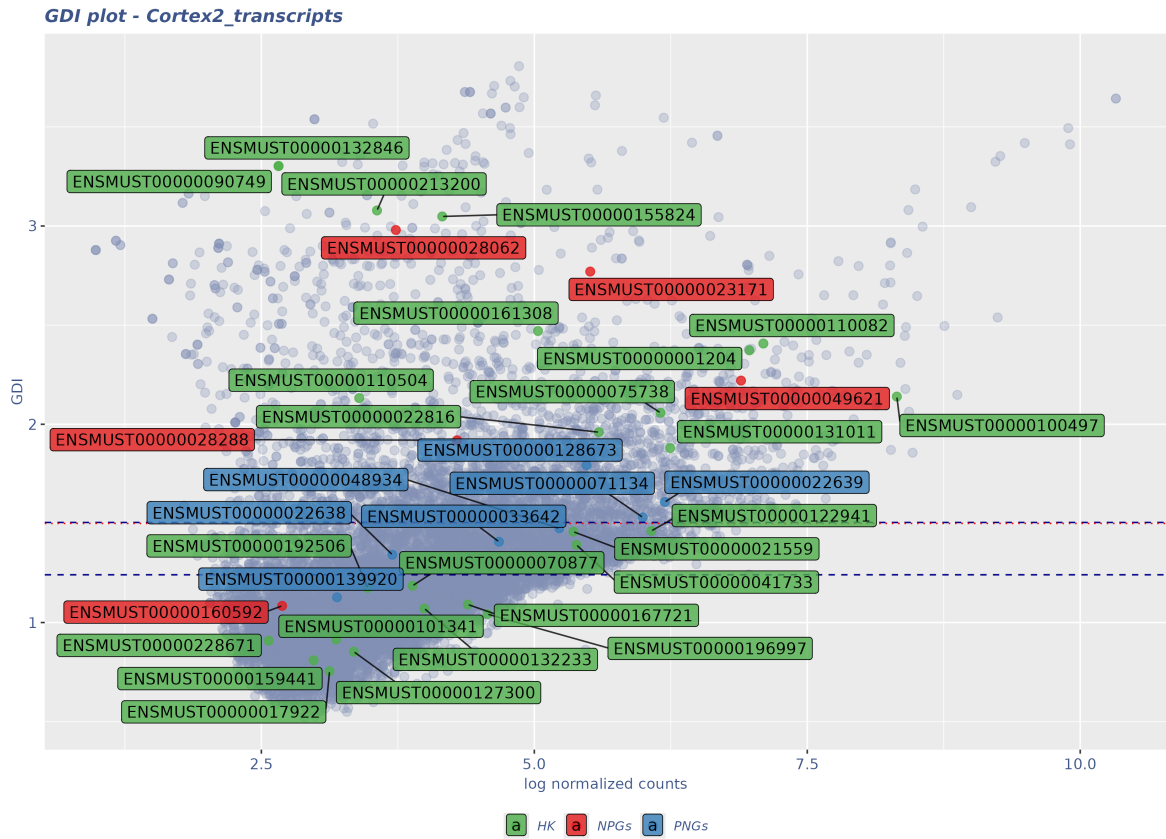
Figure 4.1 plots the GDI of the mouse cortex dataset computed on transcript counts. Each point represents a single transcript, with the x-axis showing log-scaled average expression and the y-axis showing the GDI value. Highlighted labels track specific transcripts belonging to known housekeeping (HK), neural-progenitor (NPG), and post-mitotic-neuron (PNG) marker genes.

Crucially, the transcript-level distribution upends classical gene-level expectations, revealing the mechanical realities of isoform quantification:

**The Minor Isoform Collapse:** Several transcripts belonging to established NPG and PNG marker genes sit unexpectedly low, hovering near the blue null median line of roughly 1.305. While the parent genes strongly identify cell types in aggregate, their minor or unexpressed isoforms fall victim to extreme sparsity. Without sufficient detection density to generate significant co-expression *p*-values, these specific variants flatline. This confirms that a marker gene’s predictive power does not automatically transfer to all its constituent transcripts.

**The Housekeeping Variance Spike:** Conversely, several HK transcripts break well above the red target threshold of 1.5. Because housekeeping genes are universally highly expressed, they generate massive quantities of 3’-biased terminal reads. The pseudo-aligner’s Expectation-Maximization (EM) algorithm must distribute these ambiguous reads across overlapping equivalence classes. This process can introduce highly variable, cluster-correlated fractional assignments that COTAN correctly flags as structural variance, exposing either an EM quantification artifact or a genuine, unexpected isoform switch within a constitutive gene.

**Decoupling Gene from Transcript:** The presence of a broad cloud of transcripts rising well above the threshold confirms that the sparse matrix retains robust differential signal. However, the inversion of the highlighted markers demonstrates that transcript-level variation is fundamentally decoupled from parent-gene identity, validating the necessity of analyzing these features as independent statistical entities.



**Figure 4.1:** Transcript-level GDI of the mouse cortex dataset: one point per transcript, highlighted markers are housekeeping (HK), neural-progenitor (NPG), and post-mitotic-neuron (PNG) transcripts. Null median at roughly 1.305; target threshold at 1.5.

## Gene vs Transcript GDI Comparison

Figure 4.2 repeats the GDI computation on the standard gene-level matrix, highlighting the identical housekeeping and marker categories. Because the gene run utilizes the denser, independent GEO matrix (28,692 features against the transcript matrix's 12,848), it serves as a baseline for how these markers behave when isoform-level sparsity is removed.

Comparing this gene-level distribution to the transcript-level plot reveals the profound impact of aggregation:

**Recovery of Marker Signal:** At the gene level, the expected biological hierarchy is restored. Neural-progenitor (NPG) and post-mitotic-neuron (PNG) markers—such as *Map2*, *Hes5*, and *Vim*—clearly detach from the baseline cloud and ascend to high GDI values (2.0 to 4.0). By summing the reads of all constituent isoforms, the matrix recovers the statistical density required for COTAN to easily separate these distinct cellular populations.

**Physiological Variance in Housekeeping Genes:** While several housekeeping genes (*Amacr*, *Tars*) anchor the baseline as expected, others (*Zfr*, *Calm1*, *Actb*) exhibit surprisingly high GDI scores. At the aggregated gene level, this spread reflects genuine biological variance. Because housekeeping genes are universally and highly expressed, their read counts are sensitive to systemic cellular differences—such as total library size, metabolic activity, and cell cycle stage—that naturally differ between a differentiating progenitor and a mature neuron. COTAN accurately captures this broad physiological variance as statistical non-independence.

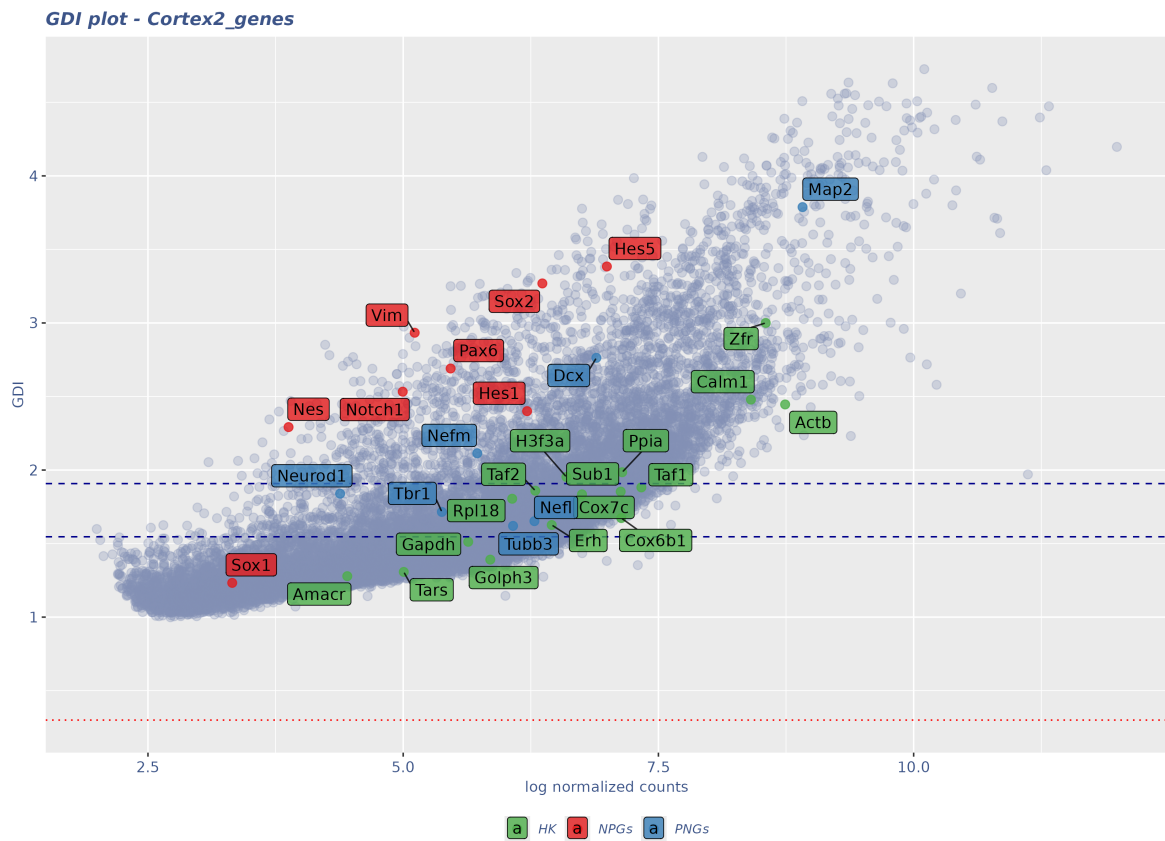
Ultimately, these two GDI analyses establish a critical foundation for the DTU pipeline: while gene-level aggregation is required to achieve the statistical power necessary to validate canonical cell-type markers, the transcript-level matrix still preserves sufficient differential signal to distinguish populations, validating the search for isoform-specific switches.

## Clustering Comparison: Gene vs Transcript Level

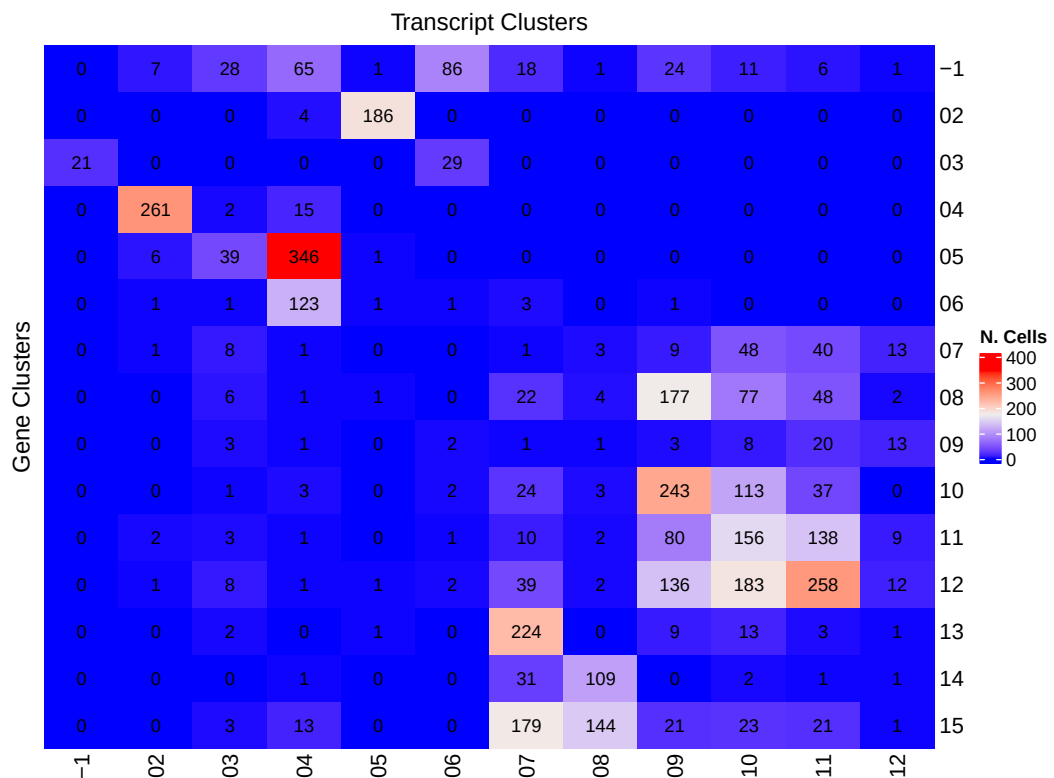
To quantify how gene and transcript clustering diverge, a confusion matrix was constructed across the 4,077 common cells present in both analyses. Figure 4.3 cross-tabulates the 11 transcript-level clusters against the 15 gene-level clusters, revealing three distinct structural patterns that justify the isoform-level resolution.

**Diagonal dominance (signal preservation):** The majority of cells receive consistent assignments across both methods, grouping into heavy blocks along the diagonal (e.g., 261 cells cleanly mapping Gene cluster 04 to Transcript cluster 02). This validates that the transcript-level pipeline does not produce random partitions: despite the extreme sparsity of the filtered transcript matrix (12,848 features versus 28,692 genes), it recovers the broad cellular architecture without being overwhelmed by technical dropouts.

**Systematic splitting (hidden sub-states):** Several broad gene-level clusters break down into finer sub-regions at the transcript level. For example, Gene cluster 15 splits cleanly into Transcript clusters 07 (179 cells) and 08 (144 cells); similarly, the massive Gene cluster 12 distributes its cells predominantly across Transcript clusters 09, 10, and 11. This divergence suggests that populations appearing homogeneous under aggregated gene expression actually contain distinct sub-groups utilising fundamentally different isoform programs.



**Figure 4.2:** GDI plot of the same dataset on gene counts, with the same housekeeping and marker categories highlighted.



**Figure 4.3:** Confusion matrix comparing transcript-level (11 clusters) and gene-level (15 clusters) assignments for the 4,077 common cells of the mouse dataset.

**Selective merging (shared splicing profiles):** Conversely, the transcript matrix collapses some distinctions enforced by the gene matrix. Transcript cluster o4 merges populations from Gene clusters o5 (346 cells) and o6 (123 cells) into a single group. Cells differing in overall transcriptional activity could nonetheless share a unified, dominant isoform-usage profile.

The discrepancy in total cluster counts (11 versus 15) is not a simple resolution ranking but a reflection of the matrices themselves: the transcript matrix is a heavily feature-filtered view, whereas the gene matrix is a denser reference count set. Ultimately, the confusion matrix demonstrates agreement in the large and divergence in the particular: the local splits and merges mark precisely where isoform information redraws biological boundaries, exposing the exact regime where downstream DTU detection—such as the transcript-exclusive *Srrm2* and *Grin2b* switches—adds biological value over standard gene-centric workflows.

However, this raises a critical question: whether the detected DTU events hold across clustering strategies, or depend entirely on how cells are partitioned. The robustness analysis below answers it.

## Core Finding: DTU Detection Results

The algorithm from Section 3.5 was applied to both human and mouse datasets, filtering for mutually exclusive isoform pairs with bidirectional cluster preference (threshold  $\tau$  as specified). The results show that contingency-table statistics can recover within-gene exclusivity patterns even in sparse 3'-biased data.

### Human Dataset Findings (Arigoni)

The human cell line mixture provided ground-truth validation through known biology. Roughly a tenth of the genes carrying at least two detected transcripts retain at least one significant mutually exclusive pair, yet after the bidirectional contrast threshold  $\tau = 0.2$  only a handful of candidates survive, across just a few genes (Table 4.3)<sup>1</sup>. That collapse is the algorithm working as designed: mutual exclusivity alone admits hundreds of pairs, but only a handful also flip their enrichment between clusters, and it is this second condition—a pair must be preferred in one cell type and disfavoured in another—that separates a genuine switch from a statistical curiosity.

| Filtering stage                              | Genes | Pairs |
|----------------------------------------------|-------|-------|
| $\geq 2$ detected transcripts                | 4,958 | —     |
| $\geq 1$ significant mutually exclusive pair | 430   | 2,006 |
| Bidirectional contrast $\tau = 0.2$          | 9     | 21    |

**Table 4.3:** Funnel of the human DTU search.

<sup>1</sup> Counted by `extract_dtu_candidates()` and recorded in the run log (`results/arrigoni/logs/`); the exported `dtu.candidates.csv` holds only the survivors.

## Mouse Dataset Findings (Ding)

For mouse brain tissue a lower threshold,  $\tau = 0.09$ , was applied, and DTU extraction was run on both clusterizations—the transcript-derived partition and the gene-derived one. Both runs start from the same funnel—14 genes carrying 53 significant pairs at the loosest point ( $\tau = 0.01$ )—but diverge once the contrast is applied: 41 candidates across 5 genes survive at  $\tau = 0.09$  in the transcript-derived clusters, and 38 across 3 genes in the gene-derived ones (Table 4.4). The inversion is itself informative—fewer initial pairs but more survivors means mouse cortex switches are gradual rather than sharp, so a relaxed threshold admits many weak but real transitions that the strict human cutoff would have discarded. The gene-derived set is a strict subset of the transcript-derived one: all 38 pairs it detects are also found by the transcript partition, which additionally captures three—Grin2b and two Srrm2 switches—exclusive to it.

| Filtering stage                                   | Genes | Pairs |
|---------------------------------------------------|-------|-------|
| $\geq 2$ detected transcripts                     | 2,122 | —     |
| $\geq 1$ significant mutually exclusive pair      | 14    | 53    |
| Bidirectional contrast $\tau = 0.09$ (transcript) | 5     | 41    |
| Bidirectional contrast $\tau = 0.09$ (gene)       | 3     | 38    |

**Table 4.4:** Funnel of the mouse DTU search.

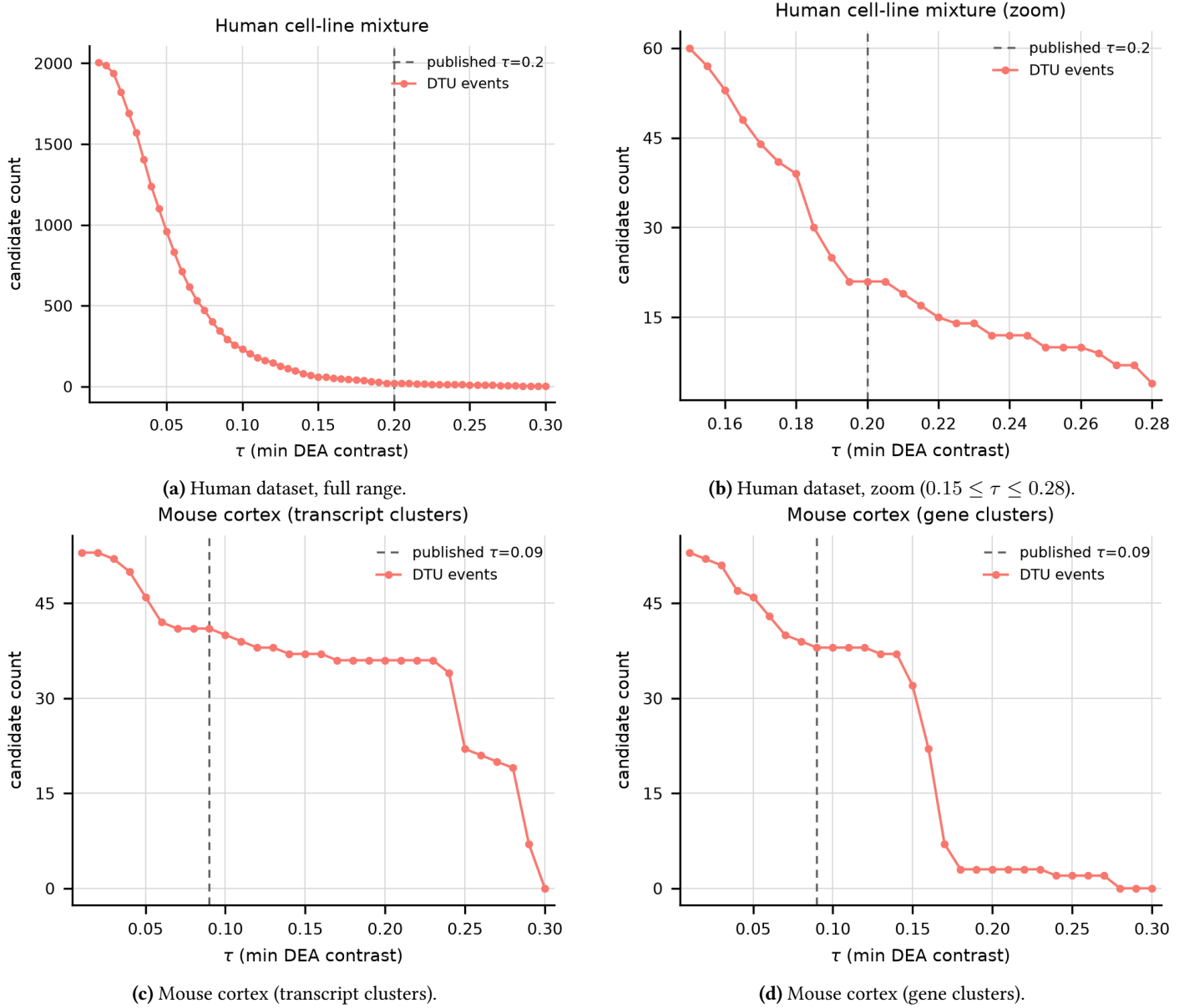
## Threshold Sensitivity: The Tau Sweep

The candidate count fundamentally depends on the differential expression contrast threshold ( $\tau$ ). Because this parameter was initially selected pragmatically to yield a biologically manageable candidate list (on the order of tens of events) rather than derived from a formal statistical null distribution, it is critical to verify that the chosen values do not sit on a fragile algorithmic edge. To evaluate this stability, the DTU extraction was re-run over a fine grid of  $\tau$  values (ranging from 0.005 to 0.300 at intervals of 0.005), recording the number of accepted events and unique genes at each step.

The resulting sensitivity sweep acts as an internal consistency check, confirming that the event count decreases monotonically as the threshold strictly increases.

**Human Dataset Dynamics (Figures 4.4a and 4.4b):** Figure 4.4a maps this threshold landscape for the homogeneous human cell-line mixture. The bulk of significant pairs descends in a smooth, continuous curve—dropping from 2006 events at  $\tau = 0.005$  to 21 events at  $\tau = 0.2$ . This nearly featureless gradient reflects the nature of the well-separated cell lines: almost every isoform pair carries some reciprocal contrast, creating a dense spectrum with no natural break.

On such a continuous slope, any single threshold might appear arbitrary. However, as shown in the zoomed-in view of Figure 4.4b, a genuine plateau emerges at  $\tau = 0.25$ , where the candidate count holds perfectly flat at ten events through  $\tau = 0.26$ . The published threshold of  $\tau = 0.2$ , which successfully reproduces the 21 reported candidates, sits immediately at the foot of this slope, just below the stable highland. Therefore, the human result is anchored at the boundary of a stable topological region rather than lost arbitrarily on a volatile tail.



**Figure 4.4:** Tau sensitivity sweeps: accepted DTU events as a function of the contrast threshold  $\tau$ .

**Mouse Dataset Dynamics:** The threshold landscape for the mouse cortex behaves structurally differently. Rather than a smooth gradient, the candidate count descends in discrete steps, with the 41 events at  $\tau = 0.09$  holding steady across the wide band  $\tau = 0.07$ – $0.09$  before stepping down through 40, 39, and 38 (Figure 4.4c). Because the heterogeneous tissue is dominated by a few high-contrast isoform switches accompanied by a highly sparse tail, its stability regions manifest as broad ledges rather than a single distinct summit. The gene-derived clusters trace a similar shape but reach extinction sooner, holding ledges at 38 ( $\tau = 0.09$ – $0.12$ ) and 37 ( $\tau = 0.13$ – $0.14$ ) before collapsing (Figure 4.4d). The chosen threshold of  $\tau = 0.09$  is the point where both partitions sit simultaneously on their plateaus—the joint-stable point that makes the call principled rather than arbitrary.

Ultimately, this parameter sweep provides a vital post-hoc anchor for the heuristic thresholds. While the selected values (0.2 for human, 0.09 for mouse) were originally targeted for validation feasibility, the sweep proves they reside within or directly adjacent to mathematically stable regions.

Consequently, the final candidate lists are robust structural findings, not artifacts of a single, fragile threshold value. However, this sensitivity sweep remains a diagnostic heuristic; establishing a true permutation-based null distribution would be the optimal future methodology to provide statistically calibrated  $p$ -values for the contrast metric.

## Discussion

### Key Findings and Literature Validation

The top candidates (Meg3 [40], Malat1 [41], TPM1 [42]) align with documented biology—lncRNA switching in neurogenesis and ribosomal protein regulation—which indicates that the algorithm recovers genuine signal rather than technical artifacts. Convergence across a human cell line mixture and mouse brain tissue further indicates that the method holds under different biological contexts rather than a single favourable case study. Table 4.5 frames these calls along a gradient from strong evidence to uncharted ground, and reading the results along that gradient both validates the method and marks what it cannot yet claim.

| Dataset             | Strong evidence   | Plausible, unconfirmed | Treat with caution  |
|---------------------|-------------------|------------------------|---------------------|
| Human / NSCLC lines | TPM1, RPS24, SAT1 | DSTN, TPM4, SNHG5      | RPL10, RPL35A, RPS8 |
| Mouse / cortex      | Meg3, Malat1      | Pde10a, Grin2b, Srm2   | —                   |

**Table 4.5:** Literature support for the DTU candidate genes. Strong-evidence genes are directly documented in prior isoform studies; plausible genes are consistent with related biology but not directly confirmed; caution genes lack prior DTU literature and need orthogonal validation.

**Strong evidence:** The strongest candidates (TPM1, Meg3, Malat1) are consistent with biology reported in prior isoform studies, but this thesis makes no biological claims beyond that consistency. TPM1 is a cell-type-specific tropomyosin switch, and Meg3 and Malat1 are long non-coding RNAs<sup>2</sup> whose alternative isoform usage is documented [40, 41, 42].

**Plausible but unconfirmed:** The middle tier (DSTN, TPM4, SNHG5; Pde10a, Grin2b, Srm2) presents biologically consistent patterns of alternative usage that align with related literature, though none is directly confirmed as an isoform switch in prior studies. These calls extend the method’s reach without yet anchoring it. On the mouse side, Pde10a is detected by both clusterizations, while Grin2b and Srm2 surface only in the transcript-derived partition at contrast just above the  $\tau = 0.09$  bar, so they sit at the boundary between plausible and confirmed.

**Treated with caution:** The final tier (RPL10, RPL35A, RPS8 in the human dataset) carries significant statistical switching in this matrix but no prior DTU literature. These novel candidates are exactly the calls that require orthogonal validation: confirming them would demand an orthogonal experiment such as quantitative PCR<sup>3</sup> or long-read sequencing, or a re-run on an independent dataset to see whether the same isoform switches recur. Binding validation to this tier turns the absence of ground truth into a concrete next step rather than a detached caveat. The mouse cortex retains no caution-tier candidates at the raised threshold: the weak, unanchored calls (Ttc14, Zfp941, Snrnp70, Lrrc7) all fall below the  $\tau = 0.09$  contrast bar and are dropped, which is itself the precision gain of the stricter threshold.

<sup>2</sup> **Long non-coding RNA (lncRNA):** A long RNA molecule that does not encode a protein but regulates gene expression, splicing, and development.

<sup>3</sup> **RT-PCR:** Reverse-transcription polymerase chain reaction: a lab technique that experimentally confirms whether a specific isoform is present, used as an orthogonal check on computational DTU calls.

A further conclusion concerns clustering resolution. The strongest signals survive both clusterizations—they are cluster-agnostic and validate the core functionality—while the sub-tler switches surface only in the transcript-derived partition: *Grin2b* and two *Srrm2* events are captured by the transcript clusters but absent from the gene-derived set, which at  $\tau = 0.09$  is a strict subset of the transcript one. The choice of partition therefore shapes which biology becomes visible: the strongest signals are robust, but fine-grained isoform transitions reveal themselves only under the isoform-aware resolution.

## Technical Limitations

The constraints that temper these results are grouped under one heading:

- **3'-bias invisibility:** Alternative splicing events occurring in transcript mid-regions remain undetectable when only 3' ends are sequenced. The DTU candidates found here represent a subset—those where isoforms differ at their terminal exons or 3' ends.
- **EM rounding effects:** Fractional EM assignments were rounded to integers for COTAN compatibility (Section 3.3). Low-abundance isoforms with small fractional counts may be distorted by this discretization, potentially introducing false negatives or altered contrast values.
- **Threshold sensitivity:** The choice of  $\tau$  lacks formal statistical grounding. The tau sweep (Section 4.4) anchors the threshold empirically at a stable plateau, but a permutation-based null distribution would still provide calibrated  $p$ -values rather than a consistency-based choice.

## Chapter 5

# Conclusions

While 3'-biased droplet protocols strip away the internal transcript structure usually required for isoform analysis, this thesis demonstrates that some *DTU* can still be reliably detected. By extending COTAN's sparsity-robust contingency-table framework to the transcript level, this work successfully recovered within-gene exclusivity patterns directly from sparse, terminal read counts.

### Summary of Findings

Before extracting *DTU* events, it was imperative to establish that the extreme sparsity inherent to 3'-biased droplet data did not obscure the underlying biological signal. The analysis demonstrated that the transcript-level *GDI* distribution successfully preserved enough differential variance to separate distinct cellular populations. Furthermore, transcript-level clustering recovered fine-grained cellular substructures that were entirely masked under standard gene-level aggregation partitions.

Applied to the two distinct single-cell datasets, the adapted contingency-table framework successfully returned detectable signals in both. In the human cell line mixture, the algorithm initially identified 430 genes exhibiting mutually exclusive isoform usage; after enforcing a rigorous bidirectional contrast filter, 21 high-confidence candidates survived. Similarly, in the highly heterogeneous mouse cortical tissue, 41 candidate switches passed the established lower contrast threshold in the transcript-derived partition (38 in the gene-derived one).

Crucially, the strongest *DTU* calls output by the pipeline—such as *TPM1* in the human cell mixture, alongside *Meg3* and *Malat1* in the mouse cortex—are highly consistent with documented isoform biology [41, 42]. To ensure these findings were robust rather than methodological artifacts, the candidates were cross-validated against an alternative clustering strategy. The majority of the top isoform switches survived this swap, demonstrating that the detected structural mutual exclusivity represents a genuine regulatory signal that does not depend on any specific partitioning scheme.

## Broader Implications

This work has implications across three domains:

**For computational genomics:** The success of contingency-table statistics at transcript resolution demonstrates that sparsity can be modelled directly rather than overcome (Section 2.2). This philosophy may extend to other single-cell challenges—allele-specific expression, chromatin accessibility peak co-occurrence, and multi-modal integration.

**For isoform biology:** Detecting DTU in 3'-biased data opens new analysis opportunities for existing large-scale datasets (e.g., Human Cell Atlas, Tabula Sapiens<sup>1</sup>) that were previously closed to isoform-level questions. While sensitivity is limited compared to full-length protocols, the trade-off between cell numbers and coverage may favor droplet methods when statistical power from thousands of cells outweighs per-cell depth.

**For COTAN framework development:** This thesis validates that COTAN's core design principles, set out in Section 2.2, generalize beyond gene-level analysis. Future extensions to other single-cell modalities (ATAC-seq<sup>2</sup> peak co-occurrence, protein abundance correlations in CITE-seq<sup>3</sup>) appear feasible.

## What This Work Leaves Behind

Beyond the findings, this thesis leaves a set of reusable artefacts, published together in the public repository that accompanies it.

**A quantification pipeline.** The Nextflow pipeline (Section 3.2) automates the path from raw reads to a count matrix, offering both the gene-level and the transcript-level index. Its steps run separately, so a re-run from an existing index does not repeat the download.

**An analysis library.** The `cotanisoform` R package collects the whole analysis path into documented functions: reading a count matrix, adapting it to COTAN, running the COTAN steps, extracting DTU candidates, and comparing runs. It can be installed and used on any single-cell dataset, independently of the two datasets studied here, and its logging layer lets a long run report its own progress and failures in one consistent trace.

**A reproducible analysis layer.** The driver scripts reproduce the complete analysis of a dataset from a single configuration file, one per dataset and feature level, so each reported table can be regenerated without interactive input.

**Documentation and reference outputs.** The repository documents the method and the parameters behind each published table, and it stores the published results themselves—the DTU candidate tables, tau-sweep and *GDI* plots, and the run logs—under `results/`, one directory per dataset.

<sup>1</sup> **Human Cell Atlas / Tabula Sapiens:** Large international reference atlases of single-cell transcriptomes across human tissues, relied on 3'-biased droplet data and were therefore closed to isoform-level questions.

<sup>2</sup> **ATAC-seq:** Assay for Transposase-Accessible Chromatin: maps genome regions where chromatin is open, indicating regulatory activity.

<sup>3</sup> **CITE-seq:** A protocol that simultaneously measures RNA and surface proteins in the same single cells, enabling multi-modal analysis.

## Limitations and Future Work

The constraints on these results—3'-bias invisibility, the absence of orthogonal validation, the heuristic threshold  $\tau$ , and EM rounding—are set out together with the directions they open in Section 4.5. The same underlying trade-off governs all four: sensitivity is bought with cells rather than coverage, so the value of the method grows as datasets do.

The framework established in this thesis opens several methodological extensions beyond the two datasets studied here:

**Integration with long-read data:** Combining short-read depth (thousands of cells) with PacBio/Nanopore<sup>4</sup> isoform annotation would anchor DTU calls in complete transcript sequences. This hybrid approach could validate computational predictions while scaling to population-level sample sizes.

**Trajectory-aware analysis:** Current clustering assumes discrete cell populations, but differentiation is continuous. Modeling isoform proportions along pseudotime<sup>5</sup> trajectories (e.g., using COTAN's framework adapted for regression rather than categorical comparison) could reveal dynamic switching patterns during transitions. Such an extension requires datasets suited to trajectory inference; the Ding samples were not, as the original paper states [22].

**Allele-specific expression extension:** COTAN's contingency table approach naturally extends to phased variants. Detecting allele-biased isoform usage would connect regulatory genetics (eQTLs, sQTLs)<sup>6</sup> to splicing outcomes in single cells—bridging population genomics with transcript regulation.

**Benchmarking against bulk DTU tools:** Systematic comparison with rMATS, SUPPA2, and DRIM-Seq [28, 43] on matched pseudo-bulks would contextualize sensitivity and specificity. This benchmark could identify regimes where single-cell methods add unique value versus when aggregation suffices.

On the computational side, the pipeline and the analysis remain two halves joined at the count matrix: the Nextflow workflow ends at the filtered matrix, and the R analysis is started separately. Folding the analysis into the workflow as one further step would let a single run cover the path from raw accessions to DTU candidates, which is the main engineering direction this work leaves open.

A newer quantifier points the same way, but at a coarser grain than the EM step used here. SCALPEL[44] is a complete Nextflow pipeline that would substitute the whole quantification half of this work, from raw reads to a transcript matrix. Where it diverges from this thesis is at quantification itself: instead of an identity transcript-to-gene index resolved by alevin-fry's EM, SCALPEL filters internal-priming artefacts out of the aligned reads and builds an APA-aware isoform-level matrix (Section 3.2). Its authors report higher sensitivity and specificity than existing tools on synthetic data, and predictions that agree with other tools and survive experimental validation on real data [44]. In future work, its transcript matrix would be fed into COTAN exactly as the matrix built here is, testing whether the downstream DTU calls sharpen—the contingency-table analysis would stay unchanged, and SCALPEL would simply supply a sharper, APA-aware input. That COTAN output would in turn be compared against SCALPEL's own differential-isoform-usage table, providing an independent DTU caller as an orthogonal cross-check on the contingency-table calls.

<sup>4</sup> **Long-read sequencing (PacBio, Oxford Nanopore):** Sequencing technologies that produce full-length transcript reads, giving complete isoform annotation unavailable from short 3'-biased reads.

<sup>5</sup> **Pseudotime:** A one-dimensional ordering of cells along a continuous developmental or differentiation trajectory, capturing gradual transitions rather than discrete populations.

<sup>6</sup> **eQTL / sQTL:** Genomic variants that measurably affect gene expression (eQTL) or splicing/isoform usage (sQTL); linking them to single-cell isoform switches connects population genetics with regulation.

Bootstrapped quantification variance is a second extension. Abundance-based DTU tools recover the uncertainty of the EM point estimates by bootstrapping, but the contingency-table statistics used here operate on binary presence and absence, so the variance of those point estimates is of limited relevance to the reported calls. It still matters at the margin, where an estimate error near the detection threshold can flip a cell’s detection state; recovering the variance by bootstrapping the quantification step would let a future analysis quantify that edge. It is left as future work rather than included here.

## Final Remarks

This thesis demonstrates that transcript-level COTAN can reveal regulatory programs hidden to gene-centric analyses, and that a contingency-table approach to sparse data respects the measurement process rather than imposing transformations designed for dense, normally distributed data. The framework established here—methodological rigour, computational reproducibility and alignment with known isoform switches—provides a foundation for that future work, which still requires orthogonal validation and the complementary technologies set out in Section 4.5. Whether applied to neurogenesis, cancer evolution, or developmental biology, the core question remains: *which transcripts are cells expressing, and how does that choice change across conditions?*

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# Appendix A

## Supplementary Data

### Complete DTU Results (Arigoni and Ding Datasets)

The tables below list every reported event. The same data are available as CSV in the public repository, so a reader can re-derive or re-filter them.

**Table A.1:** DTU events exclusive to transcript-level clustering (Ding dataset).

| Gene   | Tr. A              | Tr. B              | Clusters A | Clusters B | $\Delta$ A | $\Delta$ B | COEX   | <i>p</i> -value        |
|--------|--------------------|--------------------|------------|------------|------------|------------|--------|------------------------|
| Srrm2  | ENSMUST00000088621 | ENSMUST00000186961 | 10         | 09, 11     | 0.25       | 0.10       | -0.059 | $1.53 \times 10^{-04}$ |
| Srrm2  | ENSMUST00000088621 | ENSMUST00000233636 | 10         | 09, 11     | 0.26       | 0.12       | -0.059 | $1.78 \times 10^{-04}$ |
| Grin2b | ENSMUST00000053880 | ENSMUST00000125905 | 11         | 04         | 0.10       | 0.10       | -0.033 | $3.39 \times 10^{-02}$ |

*No events are exclusive to gene-level clustering at  $\tau = 0.09$ : the gene-derived candidate set is a strict subset of the transcript-derived one.*

**Table A.2:** Significant DTU events in the human dataset (Arigoni 2024), including cluster enrichment and maximum contrasts.

| Gene   | Tr. A           | Tr. B           | Clusters A          | Clusters B                   | $\Delta$ A | $\Delta$ B | COEX   | $p$ -value             |
|--------|-----------------|-----------------|---------------------|------------------------------|------------|------------|--------|------------------------|
| TPM1   | ENST00000334895 | ENST00000358278 | DV90, HCC78, HTB178 | A549, CCL-185-IG             | 0.33       | 0.61       | -0.101 | $7.72 \times 10^{-67}$ |
| RPL10  | ENST00000406022 | ENST00000458500 | A549                | HCC78, PC9                   | 0.22       | 0.46       | -0.086 | $2.78 \times 10^{-49}$ |
| RPL10  | ENST00000458500 | ENST00000491035 | HCC78, PC9          | A549, PBMCs                  | 0.46       | 0.25       | -0.084 | $6.41 \times 10^{-47}$ |
| TPM1   | ENST00000404484 | ENST00000651344 | HCC78, HTB178       | A549, CCL-185-IG             | 0.27       | 0.37       | -0.074 | $1.15 \times 10^{-36}$ |
| TPM1   | ENST00000317516 | ENST00000651344 | HCC78, HTB178       | A549, CCL-185-IG             | 0.28       | 0.36       | -0.068 | $4.41 \times 10^{-31}$ |
| RPS24  | ENST00000360830 | ENST00000465692 | DV90, HCC78         | CCL-185-IG, HTB178, PBMCs    | 0.28       | 0.30       | -0.051 | $1.76 \times 10^{-18}$ |
| TPM1   | ENST00000288398 | ENST00000334895 | A549, CCL-185-IG    | HTB178                       | 0.30       | 0.21       | -0.047 | $1.13 \times 10^{-15}$ |
| TPM1   | ENST00000560959 | ENST00000651344 | HCC78, HTB178       | A549, CCL-185-IG             | 0.22       | 0.32       | -0.044 | $8.17 \times 10^{-14}$ |
| RPS24  | ENST00000360830 | ENST00000482069 | DV90, HCC78         | CCL-185-IG, HTB178, PBMCs    | 0.28       | 0.28       | -0.040 | $8.15 \times 10^{-12}$ |
| TPM1   | ENST00000334895 | ENST00000651344 | DV90, HCC78, HTB178 | A549, CCL-185-IG             | 0.28       | 0.44       | -0.040 | $8.76 \times 10^{-12}$ |
| DSTN   | ENST00000449141 | ENST00000474024 | HTB178              | CCL-185-IG                   | 0.28       | 0.22       | -0.039 | $3.70 \times 10^{-11}$ |
| SAT1   | ENST00000379253 | ENST00000462639 | A549                | PC9                          | 0.26       | 0.27       | -0.038 | $6.62 \times 10^{-11}$ |
| TPM4   | ENST00000645471 | ENST00000646974 | CCL-185-IG          | HCC78                        | 0.32       | 0.39       | -0.034 | $5.27 \times 10^{-09}$ |
| SAT1   | ENST00000379251 | ENST00000462639 | A549                | PC9                          | 0.22       | 0.21       | -0.031 | $7.47 \times 10^{-08}$ |
| SAT1   | ENST00000379270 | ENST00000487713 | A549                | PC9                          | 0.25       | 0.25       | -0.031 | $9.35 \times 10^{-08}$ |
| RPL35A | ENST00000429437 | ENST00000642387 | PC9                 | PBMCs                        | 0.21       | 0.22       | -0.031 | $1.26 \times 10^{-07}$ |
| TPM1   | ENST00000558544 | ENST00000651344 | HCC78, HTB178       | A549, CCL-185-IG             | 0.23       | 0.33       | -0.019 | $1.04 \times 10^{-03}$ |
| TPM1   | ENST00000358278 | ENST00000558868 | A549, CCL-185-IG    | CRL5868, DV90, HCC78, HTB178 | 0.52       | 0.27       | -0.018 | $1.96 \times 10^{-03}$ |
| SNHG5  | ENST00000654795 | ENST00000664396 | CCL-185-IG          | PC9                          | 0.23       | 0.28       | -0.018 | $2.37 \times 10^{-03}$ |
| RPS8   | ENST00000372209 | ENST00000485390 | PC9                 | PBMCs                        | 0.21       | 0.29       | -0.017 | $4.19 \times 10^{-03}$ |
| TPM1   | ENST00000317516 | ENST00000358278 | DV90, HCC78, HTB178 | A549, CCL-185-IG             | 0.33       | 0.54       | -0.012 | $4.03 \times 10^{-02}$ |

**Table A.3:** Shared DTU events detected by both clustering methods (Ding dataset), including cell types and contrasts.

| Gene   | Tr. A              | Tr. B              | Clusters A             | Clusters B             | $\Delta$ A | $\Delta$ B | COEX   | $p$ -value             |
|--------|--------------------|--------------------|------------------------|------------------------|------------|------------|--------|------------------------|
| Meg3   | ENSMUST00000146701 | ENSMUST00000246047 | 07, 08, 09, 10, 11     | 02, 03, 04, 05, 06, -1 | 0.28       | 0.66       | -0.174 | $1.58 \times 10^{-28}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000146701 | 02, 03, 04, 05, 06, -1 | 07, 08, 09, 10, 11     | 0.63       | 0.28       | -0.160 | $1.65 \times 10^{-24}$ |
| Malat1 | ENSMUST00000173499 | ENSMUST00000245771 | 10, 11, 12             | 02, 04, 05, 06         | 0.27       | 0.29       | -0.104 | $2.91 \times 10^{-11}$ |
| Malat1 | ENSMUST00000173499 | ENSMUST00000245150 | 08, 09, 10, 11         | 02, 04, 05, 06         | 0.17       | 0.29       | -0.073 | $3.30 \times 10^{-06}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000247387 | 04                     | 11, 12                 | 0.34       | 0.29       | -0.043 | $5.70 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000246553 | 04                     | 11, 12                 | 0.34       | 0.29       | -0.042 | $6.88 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000245180 | 04                     | 11, 12                 | 0.33       | 0.30       | -0.042 | $7.27 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000246499 | 04                     | 11, 12                 | 0.33       | 0.29       | -0.042 | $7.40 \times 10^{-03}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000247387 | 02, 04, 06             | 11, 12                 | 0.37       | 0.29       | -0.042 | $7.96 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000246427 | 04                     | 11, 12                 | 0.33       | 0.29       | -0.041 | $8.84 \times 10^{-03}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000246553 | 02, 04, 06             | 11, 12                 | 0.37       | 0.29       | -0.041 | $9.49 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000246700 | 04                     | 11, 12                 | 0.33       | 0.29       | -0.041 | $9.51 \times 10^{-03}$ |
| Meg3   | ENSMUST00000245180 | ENSMUST00000246047 | 11, 12                 | 02, 04, 06             | 0.29       | 0.37       | -0.041 | $9.66 \times 10^{-03}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000246499 | 02, 04, 06             | 11, 12                 | 0.36       | 0.29       | -0.040 | $9.79 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000143836 | 04                     | 11, 12                 | 0.33       | 0.29       | -0.040 | $9.83 \times 10^{-03}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000245980 | 04                     | 11, 12                 | 0.33       | 0.29       | -0.040 | $1.04 \times 10^{-02}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000246427 | 02, 04                 | 11, 12                 | 0.36       | 0.29       | -0.040 | $1.15 \times 10^{-02}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000246700 | 02, 04                 | 11, 12                 | 0.36       | 0.29       | -0.039 | $1.24 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000247194 | 04                     | 11, 12                 | 0.32       | 0.29       | -0.039 | $1.26 \times 10^{-02}$ |
| Meg3   | ENSMUST00000143836 | ENSMUST00000246047 | 11, 12                 | 02, 04                 | 0.28       | 0.36       | -0.039 | $1.27 \times 10^{-02}$ |
| Meg3   | ENSMUST00000245980 | ENSMUST00000246047 | 11, 12                 | 02, 04                 | 0.28       | 0.36       | -0.039 | $1.36 \times 10^{-02}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000247194 | 04                     | 11, 12                 | 0.36       | 0.28       | -0.038 | $1.60 \times 10^{-02}$ |
| Pde10a | ENSMUST00000141877 | ENSMUST00000233652 | 07                     | 09, 10, 11             | 0.51       | 0.13       | -0.037 | $1.67 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000243087 | 04                     | 11                     | 0.32       | 0.24       | -0.034 | $3.21 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000246154 | 04                     | 11                     | 0.32       | 0.25       | -0.033 | $3.47 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000243937 | 04                     | 11                     | 0.31       | 0.25       | -0.033 | $3.79 \times 10^{-02}$ |
| Meg3   | ENSMUST00000243087 | ENSMUST00000246047 | 11                     | 04                     | 0.24       | 0.35       | -0.032 | $3.82 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000143847 | 04                     | 11                     | 0.31       | 0.24       | -0.032 | $3.98 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000247320 | 04                     | 11                     | 0.31       | 0.24       | -0.032 | $3.98 \times 10^{-02}$ |
| Meg3   | ENSMUST00000126289 | ENSMUST00000129245 | 11                     | 04                     | 0.25       | 0.31       | -0.032 | $4.08 \times 10^{-02}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000246154 | 04                     | 11                     | 0.35       | 0.24       | -0.032 | $4.12 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000247335 | 04                     | 11                     | 0.31       | 0.25       | -0.032 | $4.19 \times 10^{-02}$ |
| Meg3   | ENSMUST00000243937 | ENSMUST00000246047 | 11                     | 04                     | 0.24       | 0.35       | -0.031 | $4.45 \times 10^{-02}$ |
| Meg3   | ENSMUST00000143847 | ENSMUST00000246047 | 11                     | 04                     | 0.24       | 0.35       | -0.031 | $4.66 \times 10^{-02}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000247320 | 04                     | 11                     | 0.35       | 0.24       | -0.031 | $4.66 \times 10^{-02}$ |
| Meg3   | ENSMUST00000129245 | ENSMUST00000246462 | 04                     | 11                     | 0.31       | 0.24       | -0.031 | $4.73 \times 10^{-02}$ |
| Meg3   | ENSMUST00000126289 | ENSMUST00000246047 | 11                     | 04                     | 0.25       | 0.35       | -0.031 | $4.81 \times 10^{-02}$ |
| Meg3   | ENSMUST00000246047 | ENSMUST00000247335 | 04                     | 11                     | 0.35       | 0.24       | -0.031 | $4.89 \times 10^{-02}$ |